PART 3 – DRAWINGS AND SPECIFICATIONS

SECTION 1 – SUMMARY OF WORK

RFT No. Doc2487585402 Contract No. 20ECS-TI-02MR

SUMMARY OF THE WORK

1 Project Title

As identified in Row A.1 of the Information Sheet.

2 Project Location and Ward Number

As identified in Row A.2 of the Information Sheet.

3 Project Description

The work to be completed includes, but is not limited to the following:

Repair of deficient sidewalk, curb, existing multi-use trail, and road base; curb re-alignment; utility cut repairs; major road resurfacing; new multi-use trail construction, electrical signal upgrades, TTC bus landing upgrades; catch basin realignment.

4 Anticipated Commencement Date

The anticipated Commencement Date shall be as identified in Row C.2 of the Information Sheet.

If the Contract starts on a date other than the anticipated Commencement Date, the date for Total Performance of the Work will be revised appropriately to include the same number of Working Days.

5 Scheduled Date for Total Performance of the Work

This Contract, including all site restorations, clean up, and rectification of all known deficiencies, must be completed by the date identified in Row C.5 of the Information Sheet.

6 Construction Survey and Layout

All survey and layout required under this Contract shall be performed by the Owner.

During the course of the Contract, the Owner may instruct the Contractor to take over the requirements for Construction Survey and Layout. The Contractor shall perform the survey and layout required under this Contract in accordance with Part 3, Divisional Specifications DS-10 Construction Survey and Layout of this document provides further information on the requirements.

Measurement for Payment

For measurement purposes, the lump sum shall be prorated based on the actual Working Days remaining in the Contract. As a result, the full amount of the lump sum may not be paid.

Basis of Payment

Payment at the Contract Price shall be full compensation for all labour, Equipment and Material to do the Work.

7 Construction Staging and Phasing

The project shall be divided into the following two (2) phases:

Phase 1 –	Divided into four (4) stages) To be completed by November 16, 2020

Stage 1 -	Works on the north side of Ellesmere Road including tree removals, multi-use trail construction and repairs, catch basin removal and installation, replacement of driveways at station 5+00 and 5+20, curb and gutter replacement from approximately station 3+24 to 6+73 (approx. 360m), sidewalk replacement and construction including TTC bus platform, and pavement markings for the multi-use trail.
Stage 2 -	Sidewalk replacement on the south side of Ellesmere Road.
Stage 3 -	Electrical works at the Dormington Drive/Ellesmere Road intersection – completed simultaneously during Stages 1 and 2.
Stage 4 -	Site restoration for winter shutdown.

Phase 2 –	Divided into four (4) stages) To start on April 12, 2021 and be completed by May 15, 2021

Stage 1 -	Curb replacement and driveway replacement remaining from Phase 1.
Stage 2 -	Asphalt milling, road base repairs, base course asphalt paving, and utility adjustments on the south side of Ellesmere Road.
Stage 3 -	Asphalt milling, road base repairs, base course asphalt paving, and utility adjustments on the north side of Ellesmere Road.
Stage 4 -	Wearing course paving and permanent pavement marking installation.

8 Contract Drawings

The following drawings form part of this Contract:

Drawing No.	Title	Date

20ECS-TI-02MR-G1	Cover Sheet	May, 2020
20ECS-TI-02MR-G2	General Notes	May, 2020
20-02980-001 TO 20-02980-007	Major Road Resurfacing on Ellesmere Road from Orton Park Road to Markham Road	May, 2020
	Asbestos Removal Plan	May, 2020
T-2001	Pavement Marking Plan	June, 2020
PX0931	Existing Traffic Signal Removals	June, 2020
PX0931	Proposed Traffic Signal Layout	June, 2020

9 Standard Specifications and Standard Drawings

The following lists comprise the City of Toronto Construction Standard:

- List T1 - Standard Specifications for Roads
- List T2 - Standard Drawings for Roads
- List W1 - Standard Specifications for Sewers and Watermains
- List W2 - Standard Drawings For Sewers and Watermains
- Watermain and Sewer Rehabilitation Specifications
- Sewer Maintenance Hole Rehabilitation Standard Drawings
- Large Diameter Watermains 750 mm to 2300 mm Diameter
- Material Specifications
- Construction Specifications and Drawings for Traffic Control Devices

10 Geotechnical / Subsurface Reports

Geotechnical Investigation – Factual Report – 20ECS-TI-02MR Ellesmere Road- BRM-00604839-A0 (112) -Prepared by EXP Services Inc. – April 21, 2020

11 Permits and Approvals

The Contractor shall determine and, at its own expense, obtain permits approvals where required to carry out the Works under this Contract.

The City has obtained the following approvals:

Tree removal

12 List of Confined Spaces

- Entry to manholes, catch basins, valve chambers and underground vaults
- Trenches and/or excavations
- Utility maintenance holes

(a) The Contractor shall maintain the Work and the Site in a safe and tidy condition and free from the accumulation of waste products and debris, other than that caused by the Owner, other contractors or their employees.

(b) Before applying for Substantial Performance of the Work as provided in GC 5.5 - SUBSTANTIAL PERFORMANCE OF THE WORK, the Contractor shall remove waste products and debris, other than that resulting from the work of the Owner, other contractors or their employees, and shall leave the Site clean and suitable for use or occupancy by the Owner. The Contractor shall remove products, Construction Equipment, and Temporary Work not required for the performance of the remaining Work all to the satisfaction of the Contract Administrator and the Owner, acting reasonably.

(c) Prior to the final Application for the Payment, the Contractor shall remove any remaining products, Construction Equipment, Temporary Work, and waste products and debris, other than those resulting from the work of the Owner, other contractors or their employees.

(d) Contractor shall complete all maintenance and cleanup of the Work and Site within 24 hours of written notice from the Owner or Contractor Administrator of such. If such maintenance and cleanup is not completed within 24 hours of such written notice, the Owner shall be entitled to, or to engage others to, perform such maintenance and cleanup, at the Contractor's expense and set-off the costs thereof in accordance with GC 5.10 – OWNER'S SET-OFF.

(e) Contractor shall repair all damage to the Site caused by the Contractor's, Subcontractor's, Supplier's or Sub-subcontractor's transportation in and out of the Site within five (5) Working Days of written notice from the Owner or Contractor Administrator to repair or before final payment, whichever is earlier. If such repair is not completed within the required time period, the Owner shall be entitled to, or to engage others to, perform such repair, at the Contractor's expense and set-off the costs thereof in accordance with GC 5.10 – OWNER'S SET-OFF.

GS-3 Layout

(a) The Contract Administrator shall provide baseline and benchmark information for the general location, alignment, and elevation of the Work. The Owner shall be responsible only for the correctness of such information provided by the Contract Administrator.

(b) Where the Agreement provides for the Contractor to lay out the Work, this section GS-3(a) shall apply.

(i) Prior to commencement of construction, the Contract Administrator and the Contractor shall locate on site those property bars, baselines and benchmarks that are necessary to delineate the Site and to lay out the Work, all as shown on the Drawings.

(ii) The Contractor shall be responsible for the preservation of all property bars while the Work is in progress, except those property bars that must be removed to facilitate the Work. Any other property bars disturbed, damaged or removed by the Contractor's operations shall be replaced by an Ontario land surveyor, at the Contractor's cost.

(iii) At no extra cost to the Owner, the Contractor shall provide the Contract Administrator with such materials and devices as may be necessary to lay out the baseline and benchmarks, and as may be necessary for the inspection of the Work.

(iv) The Contractor shall provide qualified Personnel to lay out and establish all lines and grades necessary for construction. Such Personnel shall include a licensed land surveyor responsible for conducting a survey to verify the locations of all key structures. The Contractor shall notify the Contract Administrator of any layout Work carried out, so that the same may be checked by the Contract Administrator.

(v) The Contractor shall install and maintain substantial alignment markers and secondary benchmarks as may be required for the proper execution and inspection of the Work. The Contractor shall supply one copy of all alignment and grade sheets to the Contract Administrator.

(vi) The Contractor shall assume full responsibility for alignment, elevations and dimensions of each and all parts of the Work, regardless of whether the Contractor's layout Work has been checked by the Contract Administrator.

(vii) All stakes, marks and reference points shall be carefully preserved by the Contractor. In the case of their destruction or removal, for any reason, before the date of the Substantial Performance of the Work, such stakes, marks and reference points shall be replaced, to the satisfaction of the Contract Administrator, at the Contractor's cost.

(c) Where the Agreement provides for the Owner to lay out the Work, this section GS-3(c) shall apply.

(i) The Owner shall be responsible for setting out the line and setting out the required elevation of the specific parts of the Work identified in the Agreement.

(ii) The Owner shall supply a copy of the alignment and grade sheets to the Contractor to facilitate the construction of the Work in accordance with the Agreement.

(iii) The Owner shall install and maintain substantial alignment markers and secondary benchmarks as may be required for the proper execution and inspection of the Work.

(iv) All stakes, marks and reference points provided by the Owner shall be carefully preserved by the Contractor. In the case of the destruction or removal as a result of the Contractor's operations, such stakes, marks and reference points shall be replaced by the Owner at the Contractor's cost.

(v) The Contractor shall give the Owner at least 24 hours' notice before requiring levels, lines or stakes, on any portion of the Work and the Contractor shall clearly state in such notice the exact locality or localities where such are needed for use.

(vi) The Contractor must satisfy itself before commencing Work at any point as to the meaning and accuracy of all stakes and marks, and no Claim shall be considered by the Owner for or on account of any alleged inaccuracies or for any alterations subsequently rendered necessary on account of any such alleged inaccuracies, unless the Contractor notifies the Owner thereof in writing before commencing the Work.

(vii) The Contractor shall be responsible for the preservation of all property bars while the Work is in progress, except those property bars which must be removed to facilitate the Work. Any other property bars disturbed, damaged or removed by the Contractor's operations shall be replaced by an Ontario land surveyor, at the Contractor's cost.

GS-4 Site and Drainage

(a) The Contractor's sheds, site offices, toilets, other temporary structures and storage areas for material and equipment shall be grouped in a compact manner and maintained in a neat and orderly condition at all times.

(b) The Contractor shall keep all portions of the Work well, properly and efficiently drained, to at least the same degree as that of the existing drainage conditions, during construction and until the Work is completed. The Contractor shall be solely responsible for all Losses caused by, or resulting from, water backing up or flowing over, under, through, from, on or along any part of the Work or which any of the Work may cause to flow elsewhere and shall, at the Contractor's sole cost, repair such damage and without any extension of the Contract Time.

GS-5 Work Affecting the Property of Others

(a) Before Work is carried out that may affect the property or operations of any Ministry or agency of government or any Person, including a municipal corporation or any board or commission thereof, and in addition to such notices of the commencement of specified operations as are prescribed elsewhere in the Agreement, the Contractor shall give at least 48 hours' advance written notice of the date of commencement of such Ministry or agency of government or Person so affected.

GS-6 Quality Assurance and Quality Control

(a) The QA/QC Plan required by GC 3.13.2 shall be prepared and delivered to the Contract Administrator for review by the Contract Administrator and Owner within thirty (30) Days after the Effective Date and, after acceptance by the Contract Administrator and Owner, shall form a part of the Agreement.

(b) The QA/QC Plan shall:

(i) be based on the standards and requirements set out in the Agreement;

(ii) monitor, identify and rectify all non-compliance items within the Construction Schedule;

(c) The Contractor shall implement and perform the Work in accordance with, and in compliance with, the QA/QC Plan accepted by the Owner. The implementation of the QA/QC Plan may be subject to quality assurance audit and acceptance by the Contract Administrator and Owner. The Contract Administrator and the Owner may perform surveillance for compliance with the QA/QC Plan and examine the Work, wherever situate, for conformance.

GS-7 Project Controls and Reporting Requirements

(a) The Contractor shall perform the following obligations and comply with the following requirements:

(i) Maintain a Construction Schedule in accordance with Part 2, General Conditions, Clause 3.5 Construction Schedule and provide a fully updated Construction Schedule in advance of every site meeting or a minimum of 2 weeks;

Such obligations and requirements shall apply to all Work, unless otherwise specified in the Agreement.

The Owner may at any time and from time to time waive the requirement to include any particular item in any report in connection with the Work or may reduce the frequency of any report but in such event shall have the right to reinstate any item and increase the frequency of reporting to the times provided in the Agreement.

(b) For clarity, nothing in this section 7 shall relieve the Contractor from its obligation to execute the Work to completion in accordance with the Construction Schedule or other requirements of the Agreement.

GS-8 Owner Supplied Material – Not Applicable

GS-9 Traffic, Maintaining Roadways and Detours

(a) Except as otherwise noted in the Agreement, the Contractor assumes all the risks and responsibilities arising out of any traffic related obstruction encountered in the performance of the Work and any traffic conditions, including traffic conditions on any Highway or road giving access to the Site caused by such obstructions, and the Contractor shall not make any Claim against the Owner for any delay or Losses occasioned thereby.

(b) If the Agreement require the Contractor to maintain a Highway, the Contractor shall comply with all maintenance standards and other obligations under Laws relating Highways, including the *City of Toronto Act, 2006*.

(c) The Contractor shall designate an individual to be responsible for traffic control and work zone safety. The designated individual shall be a competent worker who is qualified because of knowledge, training, and experience to perform the duties, is familiar with Book 7 of the Ontario Traffic Manual and has knowledge of all potential or actual danger to workers and motorists. Prior to the commencement of construction, the Contractor shall notify the Contract Administrator of the name, address, position, cell phone, pager, and telephone numbers of the designated individual, and update as necessary.

(d) Where an existing Roadway is affected by construction, it shall, at all times, be kept open to traffic. The Contractor shall, at no additional cost to the Owner, be responsible for providing and maintaining, for the duration of the Work an alternative route for both pedestrian and vehicular traffic through the Site in accordance with the Ontario Traffic Manual, whether along the existing Highway under construction or on a detour road beside or adjacent to the Highway under construction.

(e) Subject to the prior written approval of the Contract Administrator, the Contractor may block traffic for short periods of time to facilitate construction of the Work in accordance with the Ontario Traffic Manual. Any temporary lane closures shall be kept to a minimum.

(f) The Contractor shall maintain, to the satisfaction of the Owner and the Contract Administrator, a road through the Site. The road through the Site shall include any detour constructed in accordance with the Agreement or required by the Contract Administrator. The cost of blading required to maintain the surface of such roads and detours shall be deemed to be included in the Fixed Price or Unit Price, as applicable. The Contractor shall not be required to apply de-icing chemicals or abrasives or carry out snowplowing unless otherwise specified in the Agreement.

(g) Where localized and separated sections of a Highway are affected by the Contractor's operations, the Contractor shall not be required to maintain intervening sections of that Highway until such times as these sections are located within the limits of the Highway affected by the Contractor's general operations under the Agreement. Nothing in this section shall be taken as limiting the Contractor's obligation to maintain all areas of a Highway affected by the traffic control measures undertaken in relation to the Work and to fulfill all traffic control responsibilities thereon.

(h) Where the Agreement provides for, or the Contract Administrator requires, detours at specific locations, payment for the construction of the detours, and if required, for the subsequent removal of the detours, shall be made at the Contract Prices appropriate to such Work.

(i) Where Work is discontinued for any extended period including seasonal shutdown, the Contractor shall, when directed by the Contract Administrator, open and place the Highway and detours in a passable, safe and satisfactory condition for public travel.

(j) Where the Contractor constructs a detour that is not specifically provided for in the Agreement or required by the Contract Administrator, the construction of the detour and, if required, the subsequent removal shall be performed at the Contractor's sole expense. The detour shall be constructed and maintained to structural and geometric standards approved by the Contract Administrator. Removal and site restoration shall be performed as directed by the Contract Administrator.

(k) Where, with the prior written approval of the Contract Administrator, a Highway is closed and the traffic diverted entirely off the Highway to any other Highway, the Contractor shall, at no extra cost to the Owner, supply, erect and maintain traffic control devices in accordance with the Ontario Traffic Manual.

(l) Compliance with the foregoing provisions shall in no way relieve the Contractor of obligations under GC 4.1 - PROTECTION OF WORK AND PROPERTY, dealing with the Contractor's responsibility for damage claims, except for claims arising on sections of a Highway within the Site that are being maintained by others.

GS-10 Roadway Work

(a) If at any time, in the opinion of the Contract Administrator, damage is being done or is likely to be done to any Roadway or any improvement thereon, outside the Site, by the Contractor’s vehicles or other equipment, whether licensed or unlicensed equipment, the Contractor shall, on the direction of the Contract Administrator, and at no extra cost to the Owner and without an extension in Contract Time, make changes or substitutions for such vehicles or equipment, and shall alter loadings, or in some other manner, remove the cause of such damage to the satisfaction of the Contract Administrator.

(b) The Contractor shall provide and ensure, at all times, and at no extra cost to the Owner:

(i) safe and adequate pedestrian and vehicular access;

(ii) continuity of utility services; and

(iii) access for any and all emergency response vehicles and services,

to any and all properties adjoining the Site.

GS-11 Working Drawings

(a) Working Drawings or Working Plans means any Drawings or Plans prepared by the Contractor for the execution of the Work and may, without limiting the generality thereof, include formwork, falsework and shoring plans, roadway (that part of the Highway designed or intended for use by vehicular traffic and includes the Shoulders.) protection plans, Shop Drawings, shop plans or erection diagrams.

(b) The Contractor shall arrange for the preparation of clearly identified and dated Working Drawings as called for by the Contract Documents.

(c) The Contractor shall submit Working Drawings to the Contract Administrator in accordance with an agreed upon schedule or otherwise with reasonable promptness and in orderly sequence so as to not cause delay in the Work. If either the Contractor or the Contract Administrator so requests they shall jointly prepare a schedule fixing the dates for submission and return of Working Drawings. Working Drawings shall be submitted in printed form. At the time of submission the Contractor shall notify the Contract Administrator in writing of any deviations from the Contract Documents that exist in the Working Drawings.

(d) The Contract Administrator shall review and return Working Drawings in accordance with an agreed upon schedule, or otherwise, with reasonable promptness so as not to cause delay.

(e) The Contract Administrator's review shall check for conformity with the design concept and for general arrangement only and such review shall not relieve the Contractor of responsibility for errors or omissions in the Working Drawings or of responsibility for meeting all requirements of the Contract Documents unless a deviation on the Working Drawings has been approved in writing by the Contract Administrator.

(f) The Contractor shall make any changes in Working Drawings that the Contract Administrator may require to make the Working Drawings consistent with the Contract Documents and resubmit unless otherwise directed by the Contract Administrator. When resubmitting, the Contractor shall notify the Contract Administrator in writing of any revisions other than those requested by the Contract Administrator.

(g) Work related to the Working Drawings shall not proceed until the Working Drawings have been signed and dated by the Contract Administrator and marked with the words "*Reviewed. Permission to construct granted*".

(h) The Contractor shall keep one set of the reviewed Working Drawings, marked as above, at the Site at all times.

Supplementary Specifications

1. Access to Site – GN103SS

Supplementary Specification October 2016

The Contractor shall be responsible for all re-grading of existing roads, landscaping and access routes to suit the purposes for site access. The Contractor shall also be responsible for the restoration of all existing roads, fencing, guide rails and landscaping to pre-construction conditions or better. Any damage to trees or other property caused by the Contractor's site access shall be corrected at no extra cost to the City.

Equipment and Material shall be stored in designated areas. Notwithstanding the foregoing, the Contractor shall at no extra cost to the City remove any Equipment or Material, which in the Contract Administrator's opinion, constitute a hazard to traffic or pedestrians.

The Contractor shall plan and schedule the routes of construction and delivery vehicles to, from and within the job site, so that vehicular movements are accommodated with minimum interference and interruptions to public traffic. Access routes shall be established to allow vehicles to merge with public traffic to avoid crossing traffic lanes.

The Contractor shall obtain the Contract Administrator's prior approval for the location of any construction access points. The Contract Administrator reserves the right to alter, reject or close same, as considered necessary. The Contractor shall provide suppliers of Equipment and Material with the location and proper use of the access points.

The Contractor is advised that no construction equipment or vehicles shall be permitted on any adjacent lands unless the Contractor has obtained written permission from the applicable property owners. All areas used by the Contractor for access or storage shall be restored to their original condition at no extra cost to the City.

Basis of Payment

All costs associated with this Work shall be considered incidental to all related items of Work. No separate payment shall be made.

2. Safety Cranes and Manlifts – GN107SS

Supplementary Specification October 2016

When work requiring the use of a crane, boom or similar equipment is undertaken in close proximity to hydro or transit overhead lines, regardless of line voltage, the Contractor shall give the appropriate authority 48-hours advance notice.

The authority will decide whether protection devices are required. Any charges for the protection shall be the responsibility of the Contractor.

Basis of Payment

All costs associated with this Work shall be considered incidental to all related items of Work. No separate payment shall be made.

3. Collection of Garbage – GN108SS

Supplementary Specification Sept 2017

Amendment to TS 1.20, April 2014

TS 1.20.09 MEASUREMENT FOR PAYMENT

TS 1.20.09.01 Garbage Collection

Subsection 1.20.09.01 of TS 1.20 is deleted in its entirety.

TS 1.20.10 BASIS OF PAYMENT

TS 1.20.10 Basis of Payment

Subsection 1.20.10 of TS 1.20 is deleted in its entirety and replaced with the following:

All costs associated with this Work shall be considered incidental to all related items of Work. No separate payment shall be made.

4. Geotechnical Investigations – GN110SS

Supplementary Specification October 2016

The geotechnical investigation reports are provided for general information only, and no engineering characteristics, or extent of the soil types found, are implied or suggested. The Contractor is solely responsible for the interpretation of the information contained in the report, and shall satisfy themselves as to sub-surface conditions before submitting their Bid. The City makes no assertion as to the geotechnical characteristics, including soils, road and groundwater conditions that will be encountered within the Working Area.

5. Quality Control – GN111SS

Supplementary Specification February 2018

The Contractor shall be responsible for all quality control sampling and testing of all supplied material. The Contractor shall prepare and submit a Quality Control Plan no later than 7 Days prior to the commencement of Work. The results of any tests performed shall be submitted within 3 Days after completion of the specific test being performed to the Contract Administrator.

6. Provisional Items – GN123SS

Supplementary Specification October 2016

Tender items for this Contract may be identified in the Tender Call as provisional items. The Contract Administrator may cancel provisional items at any time during the Contract.

The Contractor shall have no claim for loss of overhead or profit should the Contract Administrator decide to delete any or all provisional item(s).

7. Excavation Work – GN125SS

Supplementary Specification October 2016

It is strongly recommended that the Contractor follow the best practices with respect to any and all work requiring excavation as set out in the document titled, *Ontario Regional Common Ground Alliance Best Practices 8.0 – June 2014* ("Best Practices") which is available at www.orcga.com/Publications/Best-Practices. In the event of a conflict or inconsistency between the "Best Practices" and any other specification, provision or requirement of the Contract, the Contractor shall comply with the specification, provision or requirement of the Contract.

8. Payment of Bonds – GN130SS

Supplementary Specification July 2018

Payment for bonding, to include both Performance and Labour and Material Payment Bonds, shall be full compensation to cover all of the Work including all labour, Equipment and Material as well as any Change in the Work approved by the City and paid for from the contingency allowance.

Basis of Payment

Payment for insurance will be prorated based on the same percentage of value of Work performed for the billable period.

9. Payment for Insurance – GN131SS

Supplementary Specification July 2018

Payment for insurance shall be full compensation to cover all of the Work including all labour, Equipment and Material as well as any Change in the Work approved by the City and paid for from the contingency allowance.

Basis of Payment

Payment for insurance will be prorated based on the same percentage of value of Work performed for the billable period.

10. Dust and Mud Control – GN105SS

Supplementary Specification October 2016

The Contractor shall be responsible for taking all necessary measures required to keep the job site clean from dust and mud from construction operations and under no circumstances shall it cause inconvenience to adjacent properties and the general public. Should the Contractor fail to comply with this requirement, the Contract Administrator may, 12 hours after having given notice in writing to the Contractor, arrange for the necessary work to be undertaken. The costs of such Work shall be completed at the Contractor’s expense and at no extra cost to the City.

Where conditions are such that mud is tracked onto existing pavement or sidewalk or onto adjacent streets, the Contractor shall clean all fouled pavement and sidewalk at least daily to the satisfaction of the Contract Administrator.

Basis of Payment

All costs associated with this Work shall be considered incidental to all related items of Work. No separate payment shall be made.

11. Saw Cutting – GN106SS

Supplementary Specification October 2016

Basis of Payment

All saw cutting required to complete the work as specified in the Contract Documents shall be considered incidental to all related items of Work. No separate payment shall be made.

12. Clean Out Existing Catch Basins – SW303SS

Supplementary Specification September 2018

Prior to commencement of construction, all existing catch basins within the contract limits shall be inspected by the Contractor and the Contract Administrator and where necessary cleaned of all debris to facilitate the inspection of the existing catch basins to determine the need for any repairs or replacement by the Contractor.

The Contractor shall inspect and provide a list of catch basins requiring repairs or replacement to the Contract Administrator in a timely manner, so that the material can be made available without causing delays to the Contract. The City shall not be responsible for any delay or additional costs incurred if a catch basin is not cleaned out prior to the Work and is found to need repairs or replacement in the future.

The Contractor is advised that all debris in the catch basins shall be removed, including any material from usual storm water run-off or previous construction activities. The material removed shall be dried and mixed in equal parts with sand. The resulting material is then disposed as a non-hazardous, non-registrable solid industrial waste by the Contractor.

If it is determined that a pup catch basin does not require cleaning, no payment shall be made.

Upon completion of final restoration, the Contractor shall re-inspect all existing catch basins in the presence of the Contract Administrator. Any existing catch basin which requires cleaning, at the sole discretion of the Contract Administrator, shall be cleared of all debris at no extra cost to the City.

Meaurement for Payment

For measurement purposes, a count shall be made of the number of catch basins cleaned prior to the commencement of construction. Payment shall be paid at the completion of the post construction inspection.

Basis of Payment

Payment at the Contract Price shall be full compensation for all labour, Equipment and Material to do the Work.

13. Gutter Adjustment – RD407SS

Supplementary Specification September 2018

Where the Contractor is removing curb or curb and gutter adjacent to existing pavement which will not be removed as part of another operation, the Contractor shall saw cut full depth of 200 to 300 mm from the proposed face of curb or front of gutter and remove and dispose of the existing pavement to accommodate the reconstruction of the curb or curb and gutter. If the Contractor removes concrete pavement or base beyond the 300 mm limitation, the Contractor shall supply and install anchored hook dowels at 300 mm spacing to stabilize the gutter adjustment and at no extra cost to the City.

After the curb or curb and gutter is reconstructed, the Contractor shall supply and place the necessary granular and pavement materials to reinstate the roadway to its original composition. Where the curb or gutter is adjacent to composite or concrete pavement, the Contractor shall install a keyway in the concrete curb or gutter.

Temporary restoration of the gutter area to facilitate the maintenance of traffic during construction shall be considered incidental to the appropriate items. No separate payment shall be made.

Measurement of Payment

Measurement of the new gutter adjustment shall be by length in metres along the face of the curb.

Basis of Payment

Payment at the Contract Price shall be full compensation for all labour, Equipment and Material to do the Work.

14. Street Furniture – RD409SS

Supplementary Specification April 2017

For the temporary removal of non-electric powered street furniture elements such as benches, waste receptacles, newspaper corrals, publication structures, postering structures, non-ad info pillars and bicycle locking rings, the Contractor shall make a request 3-4 weeks in advance and in writing to the Contract Administrator stating the location and appurtenances to be removed. Verbal requests shall not be considered.

For the temporary removal of electric powered street furniture elements such as bus shelters and ad info pillars, the Contractor shall make a request 6 weeks in advance and in writing to the Contract Administrator stating the location and appurtenances to be removed. Verbal requests shall not be considered.

The Contract Administrator shall send removal requests to streetfurniture@toronto.ca.

Street furniture such as pay-and-display machines, OMG bins, mailboxes, newspaper boxes and bus shelters shall be removed by other forces. The Contractor shall not perform this task.

15. Remove and Dispose of Pavers – RD412SS

Supplementary Specification September 2017

Interlocking stone, unit pavers, precast slabs, granite setts and flagstone within the existing boulevard identified by the Contract Administrator, as not being reinstalled shall first be offered to the property owner. If directed by the Contract Administrator that the property owner will accept salvaged material, the Contractor is to remove and salvage interlocking stone, unit pavers, precast slabs, granite sets and flagstone including removal and disposal of any concrete base off site. The Contractor must exercise care when removing the interlocking stones, unit pavers, precast slabs, granite setts, flagstone so as not to cause any damages. Any damaged or stolen interlocking stones, unit pavers, precast slabs, granite sets and flagstone shall be replaced with identical items by the Contractor at no extra cost to the City. If refused by the property owner, the Contractor shall dispose of interlocking stones, unit pavers, precast slabs, granite sets and flagstone off-site.

Measurement for Payment

Measurement of the pavers shall be by area in square metres (m^2).

Basis of Payment

Payment at the Contract Price shall be full compensation for all labour, Equipment and Material to do the Work.

16. Cold Weather Work – RD414SS

Supplementary Specification October 2016

Based on the schedule for the Work of this Contract, the Contractor is instructed that cold weather precautions may be required according to OPSS.MUNI 904 – *Construction Specification for Concrete Structures*.

Basis of Payment

All costs associated with this work shall be considered incidental to all related items of Work. No separate payment shall be made.

17. Astral Media Bus Shelter Bases – RD415SS

Supplementary Specification September 2018

Bus Shelter with Advertising Installations

The Contractor shall box out the shelter pad area for ad bus shelter according to City and Astral Media Contract Drawings. The Contractor shall run the conduit from the nearest power source to the shelter box out area and asphalt the shelter box out area temporarily to make it safe until Astral Media can install the bus shelter.

Bus Shelter without Advertising Installations

The Contractor shall provide the concrete base for bus shelter without advertising as specified in the Contract Documents.

Inspections by Astral Media

The Contractor shall arrange for a site meeting with an Astral Media representative, who will provide the locations of the conduit connection points, location of the hand well (if required) and the routing of the electrical conduit. For scheduling purposes, the Contractor shall provide the Astral Media representative with a minimum of 14 Days advance notice of the requirement of all site inspections or visits or both.

The Contractor shall coordinate with Astral Media and the City when they plan to lay conduit. Astral Media to take photos of conduit placement for their records.

Measurement for Payment

Where separate tender items for the installation of the electrical conduit and hand wells associated with this Work are provided in the Contract Price, the actual installed quantities will be measured and payment will be made at the unit rates for these items.

Where separate tender items for the installation of the electrical conduit and hand wells associated with this Work are not provided in the Contract Price, all costs associated with the installation of these items should be considered included in the unit rate provided for the construction of the base.

Basis of Payment

Payment at the Contract Price shall be full compensation for all labour, Equipment and Material to do the Work.

18. Reduction of Volatile Organic Compound (VOC) Emissions – RD423SS

Supplementary Specification	September 2017

Environment and Climate Change Canada released a *Code of Practice for the Reduction of Volatile Organic Compound (VOC) Emissions from Cutback and Emulsified Asphalt* in February 2017. The objective is to reduce VOC emissions and generating environmental and health benefits from reducing the intensity and frequency of smog events.

The code of practice details recommended practices regarding the use to cutback and emulsified asphalt during the ozone season, record keeping and reporting. TS 3.20 – *Construction Specification for Tack Coat* is modified to reflect these new code practices.

For more detailed information and to download a copy go to: http://www.ec.gc.ca/cov-voc/default.asp?lang=En&n=05CE2B41-1.

19. Construct Box-outs – RD425SS

Supplementary Specification	September 2018

Box-outs are required for new sidewalk or boulevard areas where existing street lights, traffic signal poles or bus shelters are to remain on a temporary basis.

The box-outs shall be the same dimensions as the adjacent new sidewalk so that the edges of the box-outs match the joints of the new sidewalk. The box-outs shall be constructed with a minimum of 100 mm compacted Granular A Native or Granular A RCM and topped with a minimum of 50 mm of compacted SP 4.75, Traffic Category B, PG 58-28 asphalt mix.

Any other temporary box-out that is required as a result of the Contractor's staging plan shall be at no extra cost to the City.

Measurement for Payment

For measurement purposes, a count shall be made of the number of box-outs completed. Payment shall be paid at the completion of the post construction inspection.

Basis for Payment

Payment at the Contract Price shall be full compensation for all labour, Equipment and Material to do the Work. The Work includes the removal and disposal of all material, the supplying and placing of the granular and asphalt material and the removal of the temporary granular and asphalt material, if required.

Special Provisions

20. Capital Improvement Project Construction Sign – GN101SP

Special Provision September 2017

The Contractor is responsible for the fabrication, installation, relocation, maintenance and removal of the project construction sign. The signs must be in place prior to the commencement of the Work. Signs should be located near entry and exit points or at the beginning and end locations of the project. Signs should be posted in the area where the work is taking place and visible to the public. The Contractor shall in consultation with the Contract Administrator, determine the appropriate location taking into account obstructions and sight lines.

Upon the completion of the project, the Contractor shall remove and dispose of the signs.

The following information shall be shown on the construction sign and shall be specialized for each project location:

Field: Description	Text
1: Project Title	*Major Road Resurfacing on Ellesmere Road from Orton Park Road to Markham Road*
2: Project Details	*Road resurfacing including the replacement of deficient*

	curbs, sidewalk, existing multi-use trail and construction of new multi-use trail
3: Start Date	*October 2020*
4: End Date	*May 2021*
5: Contract Number	*20ECS-TI-02MR*

The size of the construction sign shall be 1200 mm x 1200 mm

The design layout and specifications for the Capital Construction / Improvement Sign can be found by going to Engineering and Construction Services web page at www.toronto.ca/ecs-standards.

Measurement for Payment

For measurement purposes, a count shall be made of the number of new capital improvement project construction signs fabricated and installed.

Basis of Payment

Payment at the Contract Price shall be full compensation for all labour, Equipment and Material to do the Work.

21. Pre-Construction Photos and Videos – GN114SP

Special Provision September 2018

The Contractor shall provide the Contract Administrator photos or videos or both in digital format that clearly shows every property adjacent to the Working Area prior to the start of construction. Driveways, gardens, walls, fences, culverts, walkways, trees affected by the construction should be clearly visible on the recording media.

The digital photos or video or both shall be used by the Contract Administrator to verify that the restoration of any affected areas is restored to pre-construction conditions. A written summary shall be provided with details as to when the Work was completed, which locations inspected, all photos and video logs labelled along with a map to indicate areas inspected for each street.

Should the Contractor not have pre-construction photos or video of the area being restored, shall in no way, relieve the Contractor of any responsibility to complete the restoration to match existing conditions to the complete satisfaction of the Contract Administrator and the property owners.

Measurement of Payment

Measurement for providing photos or videos or both shall be by lump sum.

Basis of Payment

Payment at the Contract Price shall be full compensation for all labour, Equipment and Material to do the Work.

22. Opening and Adjustment of Toronto Hydro Hand Wells – GN115SP

Special Provision — October 2016

Adjustments to composite Toronto Hydro Electric Services (THES) hand wells shall be made using the appropriate composite spacers. The supplier and part number for the hand wells and spacers are as follows:

Composite Power Group
Phone: 519-942-8485 x121
Pencell Plastics No. PEMR-1212-24PC-THES

The opening or adjustment or both of THES appurtenances shall be completed by crews, which have been approved by THES. Furthermore, any operation which involves opening or adjustment or both of THES hand wells, or other THES appurtenances, requires that a certified electrician be present during the operation.

Basis of Payment

All costs associated with this Work shall be considered incidental to all related items of Work. No separate payment shall be made.

23. Portable Changeable Message Sign – GN127SP

Special Provision — September 2018

Portable changeable message signs are proposed at the following locations:

Portable changeable message signs are proposed at the following locations:

Street Name	at Street Name	Display Panel Size

Ellesmere Road	Westbound, East of Orton Park Rd.	2300mm x 1400mm
Ellesmere Road	Eastbound, West of Dolly Varden Blvd.	2300mm x 1400mm
Orton Park Road	Northbound, South of Brimorton Dr.	2300mm x 1400mm
Scarborough Golf Club Road	Northbound, South of Painted Post Dr.	2300mm x 1400mm

The Contractor shall, in consultation with the Contract Administrator, determine the exact location taking into account obstructions, sight lines and right of way limitations. The Contract Administrator may request that the Portable Changeable Message Signs (PCMS) be relocated a maximum of three (3) times each at no extra cost to the City. Upon request, the Contractor must relocate the PCMS sign within 12 hours of being notified.

SCOPE

This special provision details the requirements for trailer mounted Portable Changeable Message Signs (PCMS). The Contractor shall supply and place the signs one week in advance of the Work.

REFERENCES

This Special Provision refers to the following standards, specifications or publications:

National Transportation Communications for ITS Protocol (NTCIP):

NTCIP 1203 v03 Dynamic Message Signs

DEFINITIONS

For the purpose of this Special Provision, the following definitions apply:

Pixel: means an assembly of LEDs that collectively form an image-forming unit. All LEDs in a pixel are turned on or off in unison.

PCMS: means trailer mounted, Portable Changeable Message Sign that includes sign structure, signcase, display elements, photocell sensor, CMS controller, and all other associated mechanisms and equipment required to form an operational display.

EQUIPMENT

Display

The PCMS display panel shall be small (urban) full matrix display meeting the following dimensions:

Width: 1900 to 2300 mm
Height: 1000 to 1400 mm

Larger PCMS units may be used if the desired location is suitable and approved by the Contract Administrator.

The PCMS shall incorporate a full matrix display of a minimum 30 x 55 pixels with 4 LEDs per pixel. The PCMS shall be capable of displaying a minimum of 12 characters per line at a minimum of three lines per panel.

The PCMS display shall be capable of rotating 360 degrees for multiple display options.

The PCMS shall incorporate a photo sensor system to provide automatic control of the display luminance as a function of the ambient illumination level. Messages displayed on the PCMS shall be legible from 50 to 300 metres in all ambient light conditions.

Signcase

The display elements and associated electronics shall be housed in weather-tight aluminum housing, designed to provide protection from solar radiation, water, dust, dirt and salt spray.

The sign face shall be constructed of non-glare, scratch resistant, high impact, ultraviolet radiation stabilized, polycarbonate sheeting. The polycarbonate sheets shall be hinged and prop rod secured to allow easy access to internal sign components for service and repair.

Trailer Mounting

The maximum dimensions of the Portable Changeable Message Sign while in display mode shall be as follows:

Maximum overall operating height = 4.3 m
Maximum overall width = 2.0 m
Maximum overall length = 3.8 m

Transportation

The PCMS shall be designed for towing at highway speeds. The PCMS shall incorporate a 50mm coupler or 75 mm pintle eye for easy towing and shall be adjustable in height from ground level.

Security

Wheel locks and other measures shall be provided to maintain the security of the PCMS on site. The sign case, battery enclosures and controller enclosures shall each be securely locked. Three (3) sets of master keys are to be provided for all locked components. The trailer should have a provision for securing it with a permanent structure or fixed object by a chain and lock. The PCMS shall be supplied with a chain and padlock.

The enclosure shall allow storage area for the chain/padlocks/wheel locks, etc. to be stored when not in use.

Lighting System

A 12 VDC lighting system shall be provided for the PCMS trailer and shall include tail lights, stop lights, clearance markers and license plate lights.

Paint

The PCMS and trailer assembly shall be painted construction orange.

Electrical System

All electrical/electronic components shall be of modular, interchangeable, plug-in type fabrication and shall be standard manufacturers' components and CSA certified, where possible. If no CSA standards are available for a proposed component, other standards organization certification may be substituted with the approval of the Construction Administrator.

The PCMS shall be designed to continuously operate on solar power and deep cycle batteries such that there is no requirement for external charging. The fully charged deep cycle battery pack shall be capable of powering the PCMS for 21 consecutive days (with 24 hour operation, full intensity, and at 50% pixel operation) without charging from the solar panel or any external source.

Local Control Hardware

The local controller for the PCMS shall be fully NTCIP compliant and shall be powered by the 12V DC battery system. The local controller shall have a high contrast LCD display screen and a standard 84 keyboard with a flexible waterproof cover or a high contrast or colour LCD pressure-sensitive display screen or a hand held controller capable of quick programming.

PCMS Software

All message operations for the PCMS are to be controlled locally.

Construction

The signs shall provide up-to-date information regarding the Work, road closures and delays to the travelling public. The message shall be reviewed and updated on a daily basis in order to provide accurate information to the public. The wording of all messages on each sign shall be approved by the Contract Administrator.

Measurement for Payment

For measurement purposes, a count shall be made of the maximum number of PCMS units required to be in operation at any one time during the Contract Time.

Basis of Payment

Payment at the Contract Price shall be full compensation for all labour, Equipment and Material to do the Work. Payment shall include supply, place, operate, maintain, update message board, relocate and removal of the PCMS units.

24. Sewer and Sewer Service Installation – SW301SP

Special Provision	September 2018

Installation

The installation of sewer pipe, maintenance holes, catch basins, catch basin leads and sewer laterals shall include:

- excavation, sheathing, shoring, and trench dewatering;
- the removal and disposal off site of asphalt pavement, concrete road base, brick gutter, concrete or asphalt curb, curb and gutter, monolithic curb and sidewalk, sidewalk, crosswalks, walkways, flag stone, boulevards, driveways and entrances (all thicknesses);
- remove and salvage existing unit pavers and interlock stone;
- protecting and supporting of adjacent components that shall remain during and after the work including, existing structures, sewers, laterals and watermains, services and utilities, light poles, hydro and traffic signal poles, hand wells, maintenance holes, catch basins, retaining walls, curbs, road base, driveways, sidewalks, crosswalks, walkways, valve boxes, signs and posts, fences, landscaping walls, bollards, gardens, decorative stones, garden edging and vegetation, residential sprinkler systems and lighting systems, any damage to existing components shall be repaired at no extra cost to the City;

- removal and disposal off-site of trench excavation including unshrinkable fill or concrete bedding according to TS 2.10 – Construction Specification for General Excavation;
- removal and disposal of existing sewer pipe, maintenance holes, catch basins and other appurtenances where indicated on the Contract Drawings or encountered during installation of new sewers or both;
- control of sewage flow during construction with bypass pumping shall be according to TS 4.01 – Construction Specification for Sewer Bypass Flow Pumping;
- supply and installation of sewer pipe, maintenance holes, catch basins and catch basin leads complete with gaskets and fittings;
- supply and installation of sewer laterals by open cut or augering;
- supply and installation of all tees for services, couplings, adopters, Kor-N-Seals and all components and pieces of pipes in various locations;
- connect service connections to new sewers according to TS 410 – Amendment to OPSS 410 (Nov 2012) – Construction Specification for Pipe Sewer Installation in Open Cut;
- coring into existing maintenance holes, catch basins and existing sewers shall be according to TS 410 – Amendment to OPSS 410 (Nov 2012) – Construction Specification for Pipe Sewer Installation in Open Cut;
- connections to maintenance holes, catch basins and sewers shall be by core drilling;
- connecting sewer laterals to existing sewers shall be with vertical bends;
- supply and placement of embedment or bedding material, cover material and backfill as specified in the Contract Documents (**Note**: Where Granular A material is specified in this special provision, it shall be from a source that is free of reclaimed asphalt pavement (RAP) and according to TS 1010 (April 2015) – Amendment to OPSS.MUNI 1010 – Material Specification for Aggregates – Base, Subbase, Select Subgrade and Backfill Material (April 2013));
- supply and placement of unshrinkable fill according to TS 13.10 – Construction Specification for Unshrinkable Fill;
- the installation and maintenance of permanent restoration for concrete and asphalt sidewalks, curb and roadway; and

- permanent restoration of all existing landscaping, sod, fences, river stone, gardens, decorative rocks, vegetation, interlock stone or lock stone, unit pavers, concrete and asphalt walkways or driveways (all types), to pre-construction conditions or better.

Flexible Pipe

Embedment material as it relates to flexible pipe from the bottom of the trench to the bottom of the backfill shall be Granular A Native or Granular A RCM according to TS 1010 (April 2015) – *Amendment to OPSS.MUNI 1010 – Material Specification for Aggregates - Base, Subbase, Select Subgrade, and Backfill Material (April 2013)*. The embedment material shall extend a minimum 200 mm below the pipe invert and extend 300 mm above the top of the pipe. The material shall be worked carefully under the sides of the pipe and compacted in 150 mm layers to 98% of maximum dry density with light equipment by hand so as not to damage the pipe or alter its installation in anyway. Recycled granular material shall not be used for embedment material.

The embedment and backfill shall be according to OPSD 802.010 Type 3 soil.

Rigid Pipe

Bedding material used to support rigid pipe shall be Granular A Native or Granular A RCM according to TS 1010 (April 2015) – Amendment to OPSS.MUNI 1010 – *Material Specification for Aggregates - Base, Subbase, Select Subgrade, and Backfill Material (April 2013)*. The bedding material shall be compacted by approved mechanical means in 150 mm layers to 98% of maximum dry density.

The pipe bedding, cover and backfill shall be according to OPSD 802.031 Class B bedding.

Cover Material

Cover material placed from the top of the bedding to the bottom of the backfill for rigid pipe shall be Granular A compacted by approved mechanical means in 150 mm layers to 95% of maximum dry density.

Backfill Material

Backfill material used above the embedment or cover material and below the lower of the subgrade or finished grade or the ground shall be unshrinkable fill when under concrete and asphalt sidewalks, curb and roadway. Backfill material in other areas other than roadway shall be Granular B Type II according to TS 1010 compacted by approved mechanical means in 150 mm layers to 98% of maximum dry density.

Trench Box

The use of trench boxes is permitted within the roadway on this Contract.

Sewer Service Installation and Connections

Prior to installing the sewer service lateral tees in the sewer mainline pipe, the Contractor shall confirm the location of all existing sewer service laterals.

During the installation of new sewer service connections, the Contractor shall ensure that the sewerage flow in the connection is free flowing, so that no sewage flows back into the houses. Should sewage backup into the houses, all damage caused shall be rectified at the Contractor's own expense. Prior to the installation of any sewer service connection, the Contractor shall ensure that the sewer service connection upstream of the new connection is clear of all debris and free flowing.

All existing dual connections (wye connections) shall be removed and two separate connections shall be installed in two separate trenches. New sanitary service connections to single family and semi-detached dwellings shall be individual service connections. No dual connections are permitted.

Minimum trench widths shall be as specified in the Contract Documents.

The Contractor shall install a cleanout for each service connection at the property line and extend the proposed sewer service lateral to meet the existing sewer lateral inside the private property. The Contractor shall co-ordinate with the affect property owners regarding this construction. The additional cost to plug and disconnect the existing lateral and install a bend where the proposed lateral meets the existing lateral shall be included in the Contract Price for the sewer service lateral installation. No separate payment shall be made.

The Contractor shall excavate the trench using non-mechanical methods where utility congestion is present. The cost for non-mechanical excavation shall be included in the cost for sewer service lateral connection installation. No separate payment shall be made.

Any existing water service requiring replacement shall be installed to maintain a minimum 2.5 m horizontal or 0.5 m vertical clearance from the sewer service lateral. No separate payment shall be made for adhering to this requirement.

Temporary Restoration

The Contractor shall complete temporary restoration of all roadways including curb, and curb and gutter upon completion of the sewer installation and backfill operation for every 100 m or at end of each Working Day of sewer construction, even before work is started on installation of the sewer services.

For more requirements on temporary and permanent trench restoration see supplementary specification GN121SS.

Measurement for Payment

Measurement for payment shall be according to TS 410.

Basis of Payment

Payment at the Contract Price shall be full compensation for all labour, Equipment and Material to do the Work.

Where the removal, disposal, backfill and restoration for the test pits are included in other tender items, the payment for the work shall be under the appropriate tender items.

25. Trench Stabilization – SW305SP

Special Provision October 2016

Should the Contractor encounter an unstable trench condition, the Contractor must immediately stop their operation, notify the Contract Administrator of the condition, and only proceed when approved to do so by the Contract Administrator.

The work of trench stabilization shall include:

1. Additional sub-excavation;

2. Supply and placement of Terrafix 270R non-woven geotextile overlapped a minimum of 600 mm;

3. 50 mm crushed aggregate according to TS 1010 to a depth of 300 mm for the full width of the trench; and

4. Disposal of excess material.

Basis of Payment

All costs associated with this work shall be considered incidental to all related items of Work. No separate payment shall be made.

26. Soft Spot Repairs – RD408SP

Special Provision September 2018

When the Contractor identifies soft areas that have developed after the main excavation operation has been completed, the Contractor shall notify the Contract Administrator immediately. The Contractor shall demarcate the width, length and depth of the soft spot for removal. Soft spots identified and locations approved by the Contract Administrator during the main excavation operation shall be considered as General Excavation.

In either case, the over excavated area shall be backfilled in the following manner:

1. If the existing ground is still unstable, the Contractor shall place geotextile fabric over the entire section. If the area has stabilized, the geotextile fabric shall not be required at this stage.
2. The over excavated area shall be backfilled with 50 mm crushed aggregate according to TS 1010 or other suitable material determined by the Contract Administrator to the subgrade elevation;
3. Geotextile fabric shall be placed across the section, the fabric material shall be according to TS 1860 – non-woven geotextile fabric class II
4. The geotextile fabric shall be immediately covered with the full depth of granular road base as specified in the Contract Documents, to provide the maximum stability possible.
5. If possible, the Contractor shall barricade off the area until such time as the pavement is placed.

Basis of Payment

Payment for soft spot removal (excavation), placement of geotextile fabric, supply and placement of granular material shall be according to the respective Contract Price.

27. Construction and Traffic Constraints – SP#1

Non-Standard Special Provision March 2020

The Contractor's schedule shall accommodate the below constraints, but is required to meet all stipulated contract completion dates:

Working hours during construction shall be restricted in accordance to the table below. No extension of Contract Time shall be permitted nor any delay claim shall be considered by the City due to adherence to these hours.

Day	Working Description
Monday to Friday	7am to 7pm
Saturday	7am to 7 pm
Sunday	9am to 7pm (upon approval of the contract administrator)
Civic/Statutory Holidays	9am to 7pm (upon approval of the contract administrator)

- Phase 1 work shall be completed by November 16, 2020.
- Phase 2 work must not start in 2020, and shall start on April 12, 2021 and be completed no later than May 15, 2021. To prepare for winter shutdown, Contractor shall complete temporary backfill and paving of the gutter adjustments, and be responsible for all temporary and interim measures to bring the roadway up to the final elevation of the surface asphalt so that all of the drainage can be directed to the catch basins without ponding. Contractor shall also be responsible for installing temporary measures to ensure pedestrian, cyclist and driver safety as required, including but not limited to temporary signage, pavement markings, fast fence, etc. All temporary measures for winter shutdown are at no additional cost to the city.
- Unless otherwise directed by the Contract Administrator, the Contractor shall limit their work zones to no more than 2 blocks *(or 300m)* at a time per crew. All work for each stage within the current zone shall be completed prior to moving to the next zone. The Contractor shall not restrict or block adjacent streets or adjacent signalized intersections, at any time.
- If the Contractor experiences significant delays that are outside of their control that prevent them from fully completing their work planned in the current year, the remaining work is to be assessed and agreed to by the Contract Administrator.
- The Contractor shall not start their equipment or work outside of the prescribed working hours without approval from the Contract Administrator. The Contractor shall make a request to the Contract Administrator and provide at least 5 Days in advance notice for

review and approval of any work that is to be undertaken outside of normal working hours or on Sunday.

- Once work begins on a phase or stage, the contractor shall ensure the work on that phase or stage is continuous and proceeds with expedition.
- The Contractor shall schedule their work to complete the Contract by the stipulated completion date. The cost of all overtime, weekend work, night work where permitted shall be included in the appropriate bid prices.
- The Contractor shall deploy multiple work crews for each type of work necessary to complete all of the Work within the stipulated contract performance period. The Contractor shall deploy multiple work crews simultaneously for extended periods throughout Work.
- Where the Contractor works past daylight hours, all provisions must be made to ensure a safe working environment including portable lights and generators, lighting equipped barriers as necessary.
- The contractor shall provide and maintain a 2 week schedule of working hours. 72 hours notice is required of changes to this schedule. The contractor will be responsible for reimbursing the City for any damages suffered, such as the cost of staff time, as a result of late schedule changes that are deemed to be in the contractor's control.
- If emergency works to be carried out arise as a result of the Contractor’s negligence or fault and the Contractor is required by the City to work on Sunday, Civic Holiday, Federal Holiday or on a Statutory Holiday in order to remedy the consequences of their negligence, the Contractor will not be compensated for such remedial works.
- All open cuts and sampling points within major traffic routes, when not under construction, are to be covered with counter-sunk steel plating with non-skid surfaces and open to traffic. Appropriate signs are to be posted advising of the presence of the plates. The plates must be secured to the pavement and be of sufficient thickness and strength to support the traffic. The plates are to be placed on a layer of burlap to avoid any excessive noise. The plates must also overlap the sides of the trench by 0.3 m. The plates shall be recessed into a 300 mm x 40 mm deep step joint and filled with a compacted layer of 40mm SP 4.75 asphalt. The cost of supplying, installing and maintaining steel plates is to be reflected in the unit bid prices of related works of the contract.

The following traffic restrictions shall apply:

- Only one-half of the road width, within the limits of each staging, shall be closed to vehicular traffic at a time. Uninterrupted pedestrian traffic on both side of the street shall be maintained at all times during construction. Pedestrians shall be separated from the construction activities with temporary fast fence or equivalent.

- The Contractor shall maintain uninterrupted TTC bus routes including WheelTrans services at all times during construction. The Contractor shall coordinate with the TTC for the temporary bus stop closures and provide adequate temporary signs and proper access ramps for public transit passengers.

- During periods of peak traffic (Monday - Friday 7am – 9am and 4pm to 6pm) all turning lanes shall be open to traffic at all signalized intersections.

- PDO is required for works at the Dormington Drive and Ellesmere Road intersection from Sta. 3+50 to Sta. 4+90, and at school laneways. If PDO is not available, works at these locations shall be completed during weekends only (from Friday after hours to Sunday). Weekend work shall not affect access to the Church at northeast corner of Dormington Drive and Ellesmere Road intersection. Contract Administrator and Work Zone Coordinator shall be notified at least 48hours prior to weekend works.

- Due to the requirement to inform the public of any impending closures of the roadway, the Contractor shall give the Contract Administrator at least 7 Day notice before any closures.

- When the contractor is not on site, all oncoming vehicular traffic must be separated by a solid 100mm wide yellow line.

- The Contractor shall have no claim for delay or any cost or expense arising out of:

  - a rejection by the City of a request to close lanes other than as provided in this Special Provision,

  - cancellation of any scheduled closure of lanes due to inclement weather or unforeseen circumstances, or

  - parking infractions due to construction.

- Milling and paving within all major intersection areas including signalized intersections shall commence on a Friday after 7:00 p.m. followed by asphalt paving within the major intersections and signalized intersections on Saturday or Sunday, depending on road base repairs.

- If when paving on a Saturday, inclement weather or otherwise, prevents the Contractor from completing the work, then the Contractor shall continue paving on the next day (Sunday) in order to complete the work.
- Lane widths shall be restricted to a minimum of 3.3 metres for lanes with a bus route or 3.0 metres otherwise and maintained during construction with "narrow lane" warning signage posted. Lanes shall be shifted and repainted if required.
- Once removed, stop bars and pedestrian crosswalk lines shall be installed within all open lanes before the contractor leaving the site. The Contractor shall ensure a pay duty officer remains on site until the pavement markings are installed.
- Notwithstanding the construction constraints, if no construction activity is occurring in an area, the area shall be restored and two 3.3m (min) lanes in that direction shall be maintained.

Should any unforeseen conditions arise the Contract Administrator will have the right to direct the Contractor to re-schedule construction in a manner that minimizes the effects thereof.

Construction Staging and Traffic Control Plan

The Contractor is to submit a detailed Construction Staging and Traffic Control Plan to the City for review and approval within seven (7) days of the award of the contract. Construction is not to commence without an approved staging and traffic control plan. Delays in the approval of the construction staging and traffic control plan will not justify a completion date extension of the contract.

The effectiveness of the plan will be reviewed during construction and may require adjustments as deemed necessary. The Contractor shall be prepared to modify their operations, if needed, at no extra cost to the City. Actual traffic requirements will be determined during construction by the Transportation Services Division.

The Contractor shall submit a separate detailed Paving and Traffic Control Plan to the City for review and approval within seven (7) days of the start of the milling operations. Milling or paving is not to commence without an approved staging and traffic control plan. Delays in the approval of the construction staging and traffic control plan will not justify a completion date extension of the contract.

Public Convenience

Good public relations shall be maintained between the City of Toronto and the public throughout the duration of the contract. Inquiries, complaints and problems are to be responded to immediately by the Contractor.

In carrying out the work, or any portion thereof, the convenience of the public must always be specially considered and provided for by the Contractor, who must not obstruct any street, thoroughfare or foot-walk, longer or to any greater extent than is absolutely necessary in the opinion of the Contract Administrator, and shall, in no case, excavate or open more of any street, roadway or place than is ordered or sanctioned by the Contract Administrator in writing.

The Contractor is to provide safe, ample and convenient means of approach and entrance to adjoining lanes, driveways, buildings and property, both for vehicles and foot passengers, wherever necessary, and for passing along all roadways and foot-walks, and for crossing the same where it is practicable to do so, both during the execution of the works as well as at other times, and for this purpose must construct and maintain, in good and serviceable condition, suitable and convenient platforms, approaches, structures, crossings or other works as necessary to maintain access.

The Contractor is to ensure that all residents and the schools have access to their properties at all times. If access will be blocked for a period of time, the Contractor must make arrangements with the property owner at least 48 hours in advance of any disruption. Particular attention will be required at night, to ensure that safe access is maintained for all property owners.

The Contractor shall minimize the open excavation left at the end of each working day to the extent that the excavation can be appropriately secured.

The Contractor shall note that prompt, courteous service to the Public, while carrying out the work, is essential for the successful completion of the Contract. However, if this service is not acceptable to the Contract Administrator, it may result in Contractor's negative performance evaluation.

Pedestrians

Facilitating pedestrian's safe and comfortable navigation of the work zones shall be Contractor's high priority. Uninterrupted pedestrian traffic on both side of the street shall be maintained at all times during construction.

Pedestrian traffic must be maintained on both sides of all roadways. Approval for a sidewalk closures (one side of street at any one time) may be granted for sidewalk work.

The Contractor shall only reconstruct the sidewalk of two corners only at any given time within an intersection. Satisfactory facilities for pedestrians crossing at corners must be provided.

Where pedestrians are required to use the curb lane of the existing road, the Contractor shall install full height sections of temporary concrete barrier at an angle not to exceed 45 degrees. The barrier shall be installed from the face of the curb to distance of 0.6 to 0.75 metres from the traffic lane to protect the pedestrians at no additional cost to the City. If the pedestrians are required to use any other lane, the barrier shall be extended from the curb to the limits of the temporary walkway.

Pedestrian traffic on the north side of Ellesmere Road between Scarborough Golf Club Road and Markham Road shall be maintained at all times. Proper pedestrian fence per OTM or equivalent shall be implemented to separate pedestrian traffic on sidewalk and the construction zone. In the areas where pedestrian traffic on sidewalk cannot be maintained, the curb lane shall be utilized for pedestrian traffic with proper measures as mentioned above.

Basis of Payment

All costs associated with this Work shall be incidental to all related items of Work. No separate payment shall be made.

28. Utility Locates – SP#2

Non-Standard Special Provision — March 2020

In addition to any other utility locating requirements contained in legislation and the Contract, the Contractor shall be responsible for obtaining satisfactory utility locates from all applicable utilities as necessary to complete the Contract Work. The Contractor shall be responsible to ensure the locate requests are of the appropriate area for the expected work in the month requested and to not request locates for work that will not be started prior to the expiration of the locate. The Contractor shall be responsible to adjust their request time appropriately for the month in which the locates are requested, knowing that during peak months, the locates can be delayed by several weeks.

Under no circumstances shall the Contractor commence any excavation prior to obtaining the locations of all utilities that may be affected by the Contract Work. The Contractor shall provide the Contract Administrator with a copy of the locate sheets supplied by the utility locators, prior to construction.

Not all utilities subscribe to Ontario One Call. It is therefore the Contractor's responsibility to obtain all necessary utility stakeouts, whether from Ontario One Call, the individual utility owners or by hiring private utility locating companies or both to ensure that all utilities are identified and located in a timely fashion. Regardless of the method(s) used to locate utilities, it is the Contractor's responsibility to ensure that all Contract Work is completed within the number of

Working Days or within the schedule outlined in the Contract. No extension of Contract Time shall be permitted for meeting the requirements of this Supplementary Specification, nor shall any delay claim be considered by the City if the Contractor fails to meet this responsibility.

Once the locates have been completed, the Contractor shall be responsible for verifying the location, size and depth of all underground services, including any abandoned services, with the applicable utility companies. The Contractor shall notify the Contract Administrator immediately in writing of any discrepancies between the information obtained by the Contractor and the information contained in the Contract Documents.

Before excavating across or along any utility or service, the Contractor shall determine its exact location and elevation. The utility or service shall be exposed by hand excavation and shall be adequately supported and/or protected before proceeding with machine excavation.

Basis of Payment

All costs associated with this Work shall be considered incidental to all related items of Work. No separate payment shall be made for costs incurred in obtaining utility locates nor shall there be any payments or extension of time provided due to delays in obtaining locates.

29. Utility Adjustment – SP#3

Supplementary Specification — September 2018

Amendment to TS 4.50, September 2017

TS 4.50.02 REFERENCES

Ontario Provincial Standard Specifications

Subsection 4.50.02 of TS 4.50 is amended by the addition of the following:

OPSS.MUNI 510 Construction Specification for Removal

TS 4.50.03 DEFINITIONS

Section 4.50.03 of TS 4.50 is amended by the addition of the following definitions:

Large Frame Utility Adjustments means adjustments to frame and covers that are not owned by the City and are in excess of 1000 mm any dimension.

Metro Frame and Cover means the large former Metro frame and covers including OPSD 402.030 for the large meter and valve chambers.

TS 4.50.05 MATERIALS

TS 4.50.05.03 Precast Adjustment Units

Subsection 4.50.05.03 of TS 4.50 is amended by the addition of the following to the title:

TS 4.50.05.03 Precast Adjustment Units

Precast Risers

Subsection 4.50.05.03 of TS 4.50 is amended by deleting the first paragraph in its entirety and replacing it with the following:

Adjustment units and risers for maintenance holes, valve chambers and catch basins shall be according to OPSS 1351.

TS 4.50.07 CONSTRUCTION

TS 4.50.07.01 General

Subsection 4.50.07.01 of TS 4.50 is amended by deleting the second paragraph in its entirety and replacing it with the following:

The adjustment of all appurtenances belonging to utility companies shall only be performed by Contractors approved by the appropriate utility company. The Contractor shall be responsible to contact the appropriate utility companies to seek approval for their subcontractors and to organize the work and make arrangements for any new components that may be required. The material supplied by the utility company shall be at no cost to the Contractor, but they may be required to pick up new components from the utility company's yard.

Subsection 4.50.07.01 of TS 4.50 is amended by deleting the fifth paragraph in its entirety and replacing it with the following:

For precast or poured in place maintenance holes, valve chambers and catch basins, adjustment units shall consist of brick and mortar or precast concrete adjustment units. The total depth of existing and new adjustment units shall not exceed 300 mm. Where the total depth exceeds 300 mm, the Contractor shall remove all adjustment units down to the top of the sound structure and install precast concrete or poured in place concrete risers. The top of the riser shall allow for 50 mm to 100 mm depth of adjustment units to be installed between the riser and the underside of the frame.

For maintenance holes, valve chambers and catch basins that are constructed of brick, the Contractor shall remove all loose and broken bricks and reinstate the brick structure using brick and mortar. Precast concrete adjustment units may only be used if directed by the Contract Administrator.

TS 4.50.07.03 Concrete Risers

Subsection 4.50.07.03 of TS 4.50 is amended by renaming the title and deleting the first paragraph in its entirety and replacing it with the following:

For poured in place risers, formwork shall be used on all sides of the extension. The top of the concrete on the existing structure shall be thoroughly cleaned and roughened to ensure a satisfactory bond.

For precast risers, the top of the concrete on the existing structure shall be thoroughly cleaned and smoothed to allow the proper installation of tape sealer.

For catch basins, lateral adjustments may be made by sloping the concrete riser to conform to the curb alignment. The slope shall be limited to 100 mm horizontal to 300 mm vertical, and the resulting opening shall not be less than 375 measured at right angles to the curb.

TS 4.50.07.04 Concrete Collars

Section 4.50.07 of TS 4.50 is amended by the addition of following subsection:

Valve frame and covers, within the roadway, whose width or diameter of the frame is less than 200 mm shall require the installation of a concrete collar to stabilize the adjustment. Enbridge Gas frame and covers are exempt from the requirement for concrete collars. All collars shall be constructed of concrete meeting the specifications of TS 1350 for road base, except that 37.5 mm aggregate is not permitted. All collars shall be formed using the appropriate sized sonutube to provide a minimum diameter of 300mm or a collar thickness of at least 75mm, whichever is greater. 10M stainless steel rebar bent into a circular shape, with a minimum overlap of 150 mm. The diameter of the rebar circle shall be so that the cover is equal on both sides—that is 150 mm cover shall have rebar of diameter 225 mm. The rebar shall be placed with a minimum of 50 mm cover, but no greater than 65 mm, from the top of the collar. All pieces of rebar shall be wired together prior to the concrete placement.

The finished surface shall be flush with the frame and cover and the surrounding asphalt surface. The frame and cover and concrete collar shall be fully protected during the Work, especially when milling and paving base course asphalt.

TS 4.50.09.01 Utility Adjustments

Subsection 4.50.09.01 of TS 4.50 is amended by deleting the subsection in its entirety and replacing it with the following:

TS 4.50.09.01 Utility Adjustments

Large Frame Utility Adjustments

For measurement purposes, a count shall be made of the number of adjustments performed, including those adjustments that required new frame and covers. The height of the adjustment shall be measured from the top of the concrete structure or remaining adjustment units to the bottom of the frame and cover. Adjustment heights of 300 mm or less will be paid at the specified rate. For changes in height of more than 300 mm, the rate will be prorated based on a height of 300 mm, including any necessary concrete risers. Measurements will be made to the nearest 10 mm increment.

All utility adjustments, except for Large Frame Utility Adjustments shall be prorated in accordance with Table 2. Large Frame Utility Adjustments shall be paid at a rate of one each.

Table 2: Adjustment distance and rate

Utility	Rate
Standard frame and covers whose width or diameter of the frame is between 500 mm and 1000 mm	1
Small frame and covers whose width or diameter of the frame is less than 500 mm without concrete collars	1/3
Small frame and covers whose width or diameter of the frame is less than 500 mm with concrete collars	1/2
Metro Frame and Covers	3

TS 4.50.09.02 Concrete Extension

Subsection 4.50.09.02 of TS 4.50 is amended by the deletion of the subsection in its entirety.

TS 4.50.09.03 New Frame and Covers

Subsection 4.50.09.03 of TS 4.50 is amended by deleting the subsection in its entirety and replacing it with the following:

TS 4.50.09.03 New Frame and Covers

New Metro Frame and Covers

For measurement purposes, a count shall be made of the number of new frame and covers installed on City owned infrastructure.

New utility frame and covers supplied by the utility companies shall be considered incidental to the adjustments. No separate measurement shall be made.

TS 4.50.10 BASIS OF PAYMENT

TS 4.50.10.01 Utility Adjustments – Item

Subsection 4.50.10.01 of TS 4.50 is amended by deleting the first paragraph in its entirety and replacing it with the following:

Payment at the Contract Price shall be full compensation for all labour, Equipment and Material to do the work. Payment shall include the removal and disposal of loose brick and other debris, the supplying and placing of brick or precast adjustment units, supply and installation of concrete risers, supply and installation of any necessary steps, the coordination of utility owners with the contract staging and the cleaning of all utility chambers, maintenance holes, valve boxes and catch basins within the contract limits.

TS 4.50.10.02 Concrete Extension – Item

Subsection 4.50.10.02 of TS 4.50 is amended by the deletion of the subsection in its entirety and replacing it with the following:

TS 4.50.10.02 Large Frame Utility Adjustments – Item

Payment at the Contract Price shall be full compensation for all labour, Equipment and Material to do the work. Payment shall include the removal and disposal of loose brick and other debris, the supplying and placing of brick or precast adjustment units, supply and installation of concrete risers, supply and installation of any necessary steps, the coordination of utility owners with the contract staging, picking up and delivering to the site any new frame and covers, completing work by approved contractors and the cleaning of all utility chambers, maintenance holes, valve boxes and catch basins within the contract limits.

No additional payment shall be made for any interim adjustments to raise or lower the appurtenance, in order to perform the work.

Adjustment of any valves, utility frame and covers within the sidewalk areas shall be considered part of the Contract Price. No separate payment shall be made.

TS 4.50.10.03 New Frame and Covers – Item

Subsection 4.50.10.03 of TS 4.50 is amended by deleting the first paragraph in its entirety and replacing it with the following:

Payment at the Contract Price shall be full compensation for all labour, Equipment and Material to do the work. **Payment for the required adjustment to the new frame and cover shall be made under the appropriate adjustment item.**

New utility frame and covers supplied by the utility companies shall be considered incidental to the adjustments. No separate payment shall be made.

30. Asphalt Testing – SP#4

Non-Standard Special Provision January 2019

All field samples, including material samples from the asphalt supplier, shall be taken in the presence of the City's material testing consultant and immediately provided to the tester to transport and conduct testing. The testing consultant shall also be in agreement with the method of sampling. In case of disagreement, the method followed shall be at the direction of the testing consultant. Any sample that is taken in the absence of or in disagreement with the City's material testing consultant shall be considered rejected. Any material supplied that is represented by a rejected sample shall be removed and replaced at the sole cost of the Contractor.

31. Notices to Public – SP#5

Non-Standard Special Provision January 2019

Construction Notice

A Construction Notice, prepared by the Contract Administrator that includes a detailed description of the upcoming construction work and indicates the name of the Contract Administrator and company name and contact phone number of the Field Ambassador, including any details of periodic interruptions of water service, details of temporary water service connections and construction activities that may impact public access to the site.

The notice shall be delivered by the Contractor to all properties within the contract area no less than Seven (7) Working Days before any activities by the Contractor.

Specialized Messaging

From time to time there may be additional notices to communicate key messages affecting individuals or a group of properties. Examples include residential or commercial or multi-residential water shut-offs, parking restrictions, driveway access, tree removals, TTC service and so on. Five (5) Working Days notice shall be given to the Contract Administrator to prepare the notice. The Contractor shall deliver the notice within two (2) Working Days.

Basis of Payment

All costs associated with this Work shall be considered incidental to all related items of Work. No separate payment shall be made.

32. Access to Properties – SP#6

Non-Standard Special Provision January 2019

Further to *General Conditions of Contract* clauses GC 7.07, GC 6.01 and GC 6.02, the Contractor shall direct their operations to minimize any inconvenience to the owners or occupants of the affected properties. The Contractor shall supply, place and compact granular material or hot mix asphalt or both to provide ramps for temporary driveway access and to provide safe and pothole-free access along roadways under construction.

In the event that in providing vehicular access, the installation of the sidewalk, driveway aprons and related road curb is carried out half driveway at a time, the Contractor shall not be compensated for any hand labour or concrete under-load charges.

Basis of Payment

All costs associated with this Work shall be considered incidental to all related items of Work. No separate payment shall be made. The Contractor shall be compensated for 'hi-early' strength concrete premiums for permanent concrete work to minimize closures, if approved by the Contract Administrator in advance of the work.

33. Restoration Work – SP#7

Non-Standard Special Provision January 2019

Restoration of Driveways, Sidewalks and Private Walkways

Driveway thicknesses for residential, industrial and commercial properties shall be according to T-310-050-8, except that the thickness of the concrete for commercial entrances shall be 200mm. For fire department access, the thickness shall be increased to 250mm. The base of driveways, sidewalks and private walks shall be restored with a 150 mm depth of Granular A or Granular A RCM according to TS 1010 compacted to 100% of maximum dry density.

All driveways and sidewalks shall be saw cut in a straight line. All asphalt or concrete driveways shall be paved for the full width of the driveway so that driveways will have only one straight joint. If short services are located within the asphalt or concrete driveway, the driveway shall be paved from the service connection pit to the curb or from the connection pit to the back of sidewalk, so that driveways will only have one straight joint.

Asphalt, concrete and interlocking stone driveways disturbed during construction shall be restored to equal condition or better. Driveway concrete curbs, including curb returns, sidewalks, retaining walls and private walkways impacted by construction shall be reconstructed to original condition or better.

The Contractor shall permanently restore driveways of all types and sidewalks from expansion joint to expansion joint—minimum three bays— that will be affected due to the underground infrastructure construction.

Permanent restoration for driveways, sidewalks and private walkways shall be completed within 3 Working Days after the completion of the underground services of each block.

Basis of Payment

All costs associated with this Work shall be included in the Contract Price for underground infrastructure installation. No separate payment shall be made.

Boulevard Sodded Areas

Within sodded boulevards, all trenches and excavations backfilled with Granular B Type II or select native material shall be restored with 100 mm topsoil and sod according to TS 5.10 – Construction Specification for Growing Medium and TS 5.00 – Construction Specification for Sodding, respectively. The Work includes the removal and disposal of an equivalent amount of materials.

Basis of Payment

All costs associated with this Work shall be included in the Contract Price for underground infrastructure construction. No separate payment shall be made.

Restoration of Landscaping Features

Where retaining walls, fences or other landscaping features including river stone, gardens, decorative rocks, vegetation, interlock stone or lock stone and unit pavers that could require restoration and are fronting any lots the Contractor shall, if machine usage is not possible, remove the existing sidewalk/boulevard by hand. If any damage occurs to the retaining wall, fence or landscaping features during construction, then the Contractor shall repair it to its original condition or better at no extra cost to the City.

Subdrains

The Contractor shall make permanent repairs to all existing subdrains to match existing conditions or better.

Basis of Payment

All costs associated with this Work shall be included in the Contract Price for underground infrastructure installation. No separate payment shall be made.

Retaining Wall

Any retaining wall or structure that could require reconstruction or replacement as a result of the installation of the sanitary sewer laterals, storm sewer laterals or water services shall be constructed to the original details, including foundation and new compacted granular backfill, to the satisfaction of both the property owner and the Contract Administrator.

The Contractor shall review all existing retaining walls to determine the work necessary prior to submitting a Bid.

Basis of Payment

All costs associated with this Work shall be included in the Contract Price for sanitary sewer laterals, storm sewer laterals or water service installation. No separate payment shall be made.

Temporary Trench Restoration

Temporary restoration of trenches within roadways, driveways, sidewalks, and intersections shall be completed within 24 hours after backfilling of the trench with unshrinkable fill according to TS 13.10 – Construction Specification for Unshrinkable Fill with an 80 mm lift of Superpave 19.0, Traffic Category B, PG 58-28 asphalt mix to the top of the existing asphalt surface for maintenance of traffic.

Basis of Payment

All costs for temporary trench restoration shall be included in the Contract Price for linear underground infrastructure installation items. No separate payment shall be made.

Permanent Trench Restoration

Within the paved roadway, all trenches and excavations shall be restored according to TS 4.60 – Construction Specification for Utility Cut and Restoration and according to the existing roadway structure as indicated in the geotechnical report and the appropriate permanent trench restoration detail as specified in the Contract Documents. The trench restoration shall include a stepped joint for both composite and flexible pavements.

The permanent asphalt trench restoration thicknesses shall be the greater of the depths shown in the Contract Documents, or match existing asphalt thickness.

The asphalt base course, concrete base, and edges of the existing pavement are to be tack coated with SS-1 emulsified asphalt. Cost of supplying and application of the tack coat shall be incidental.

The Contractor is advised that the City will not consider additional payment for the restoration of any over breaks that might occur at the edges of the trenches. This work shall be included in the Contract Price. No separate payment shall be made.

The Contractor shall permanently restore all roadways, concrete and granular road base, curbs, curb and gutters, driveways and sidewalks. For sidewalks, the restoration shall be from expansion joint to expansion joint (minimum three bays).

When restored with concrete and composite pavement, trench restoration shall include up to the top surface. The width of trench shall include the outer diameter of the pipe plus 300 mm on either side of the pipe.

Permanent restoration of roadway asphalt pavement on each street shall be completed within 14 Working Days after the completion of the underground infrastructure installation on that street unless otherwise approved by the Contract Administrator.

Basis of Payment

All costs for temporary trench restoration shall be included in the Contract Price for linear underground infrastructure installation items. No separate payment shall be made.

All costs for permanent restoration, up to the final lift of base course asphalt, shall be included in the Contract Price for linear underground infrastructure installation items. No separate payment shall be made. Separate payments shall be made for asphalt milling and the placement of top course asphalt over the trench according to the Contract Price of the respective items.

Utility adjustments and pavement markings for the permanent restoration over the trench shall be paid according to the Contract Price of the respective items.

Supplemental costs for trench restoration when asphalt containing asbestos, composite pavement over flexible pavement and concrete pavement over flexible pavement shall be paid under the respective item.

34. Payment for Mobilization and Demobilization – SP#8

Non-Standard Special Provision — January 2019

The tender item "Mobilization and Demobilization" is to encompass all costs associated with mobilization and demobilization activities required by the Contract at the time of award.

Basis of Payment

Payment for tender item Costs for Mobilization and Demobilization will be made as follows:

a) For Work that occurs within a single site, 60 per cent of the costs will be paid upon completion of mobilization provided progress of the Work can be sustained. The remaining 40 per cent of the costs will be paid upon the completion of the demobilization.

b) For Work that occurs on multiple sites, that is streets that are geographically separated by space and where a tender item is identified for each site, 60 per cent of the costs will be paid upon completion of mobilization for the respective site provided the Work starts within 4 days thereafter and is continuous through to completion. The remaining 40 per cent of the costs will be paid upon the completion of the demobilization.

c) For Work that occurs on multiple sites, that is streets that are geographically separated by space and where a single tender item is identified for the contract, the costs will be prorated based on the total bid price for each site relative to the total contract price and upon completion of mobilization for the respective site, 60 per cent of the cost will be paid provided the Work starts within 4 days thereafter and is continuous through to completion. The remaining 40 per cent of the costs will be paid upon the completion of the demobilization of the respective site.

35. General Excavation – SP#9

Non-Standard Special Provision January 2019

All items requiring excavation shall be in accordance with TS 2.10 General Excavation. The cost to excavate and dispose of material to allow for the placement of new granular, concrete, asphalt, topsoil, sod or other material shall be incidental to the associated items. No separate payment shall be made.

Amendment to TS 2.10, September 2017

TS 2.10.09 MEASUREMENT FOR PAYMENT

TS 2.10.09.01 General Excavation

Additional Excavation of Soft Spots (Provisional)

Subsection 2.10.09.01 of TS 2.10 is amended by the deletion of the entire section and replacing with the following:

Measurement shall be by volume in cubic metres (m^3). The volume shall be calculated as the difference between cross sections of the original ground and the cross sections of the final finished grade, in areas of cut according to T-216.02-9. The cost to excavate and dispose of material to allow for the placement of new granular, concrete, asphalt, topsoil, sod or other material shall be incidental to the associated items. No separate payment shall be made. The volume of excavated material may be as measured and agreed upon in the field if the average end area is impracticable.

Where a separate item is included for removal of concrete, asphalt, topsoil, sod or other material within the excavation limits, the volume shall not be deducted from measured volume of general excavation. Excavation beyond the lines and grades of the proposed cross sections will be disallowed unless authorized by the Contract Administrator in writing. Field cross sections shall be taken, by the Contractor, to establish the actual lines and grades where they differ from the proposed cross sections.

36. Driveways and Asphalt Boulevards – SP#10

Non-Standard Special Provision January 2019

Unless otherwise stated in the tender documents, the asphalt driveway and boulevard items shall be constructed according to specification TS 3.30 and drawing T-310.050-8. Asphalt sidewalk and boulevard shall be constructed as a residential asphalt driveway.

Unless otherwise stated in the tender documents, the concrete driveways shall be constructed according to specification TS 3.70 for concrete sidewalks and drawing T-310.050-8, except that commercial concrete driveways shall be 200mm thick. For fire department access, the thickness shall be increased to 250mm.

37. Milling and Paving – SP #11

Non-Standard Special Provision March 2020

In general, all pavement under this project will be milled up to 100 mm to allow for the placement of 50 mm of base course and 40 mm of surface course to the desired profile. The intent is not to make major changes to the existing profile, but to improve the crossfall where necessary. Any areas that require a major change to the profile by grinding either more than 100 mm or less than 70 mm shall be reviewed and approved by the Contract Administrator (Change Directive) prior to grinding.

1) **Deep milling**

Before, during or after the milling operations, the contract administrator may identify areas where an additional 50 mm shall be milled to address pavement distresses. To allow adequate cooling time, the deep milled area shall be paved level with the surrounding milled surface with base course asphalt prior to paving the rest of the roadway. Isolated road settlement along curb and gutter on Ellesmere Road has been observed at multiple locations within the project area. Additional 50mm milling may be required within approximately 1.2m margin along the curb gutter at the settled areas depending on the condition of the underlining asphalt. Decision will be made by the Contract Administrator after reviewing the underlying asphalt.

Basis for Payment

Payment will be made by prorating the appropriate cold milling contract item to the item's 100 mm mill thickness. Example: 90 mm plus an additional 50 mm =140 mm. Prorate depth over 100 mm, thus [140 mm] / [100 mm] x [unit price per m^2].

2) **Profiling**

In areas where profiling is required to achieve adequate crossfall and/or to correct undulations, the profiling shall be achieved during the milling process.

Basis for Payment

All costs associated with this work shall be considered incidental to all related items of work. No separate payment shall be made.

3) **Surface Preparation**

The surface preparation shall be approved prior to the commencement of paving operations. As a minimum, the surface shall be mechanically swept and tack coat applied as per TS 3.20.

Basis for Payment

All costs associated with this work shall be considered incidental to all related items of work. No separate payment shall be made.

4) **Paving**

Paving shall be in echelon for all surface course asphalt operations and where conditions permit for base course asphalt.

A material transfer vehicle "shuttle buggy" shall be utilized for every paving operation.

Screed vibration must be active at all times and at the proper level for the material being placed. Asphalt placed without the proper vibration will be rejected.

Tools, boots, etc. shall be cleaned using an approved release agent, such as soap/detergent solution. The use of kerosene, diesel or other solvent based release agents will not be permitted.

Basis for Payment

All costs associated with this work shall be considered incidental to all related items of work. No separate payment shall be made.

Amendment to TS 310, September 2018

TS 310.09 MEASUREMENT FOR PAYMENT

TS 310.09.01.03 Payment Mass

Subsection 310.09.01.03 of TS 310 is amended by the deleting the section in its entirety and replacing it with the following:

When the Pricing Form specifies the thickness of hot mix asphalt course(s) and that the hot mix item(s) shall be measured by mass, the actual mass of hot mix used to produce the course shall not exceed the theoretical mass by more than 10 per cent for base course asphalt and 5 per cent for surface course asphalt.

If the mass of hot mix type placed is equal to or less than the specified limit, the payment mass will be the mass of hot mix actually placed as recorded by the weigh tickets, multiplied by the density factor for the hot mix type and composition given in Table 6, as applicable.

If the mass of hot mix type placed exceeds the specified limit of the theoretical mass for the hot mix type, the payment mass will be based on no more than specified limit of the theoretical mass, multiplied by the density factor for the hot mix type and composition given in Table 6, as applicable.

38. Cold Wet Mill and Disposal of Asphalt Pavement Containing Asbestos – SP#12

Non-Standard Special Provision March 2020

The asphalt pavement on the following road sections are identified in the geotechnical report in the Contract Documents to contain 0.5 per cent or more of asbestos.

Street Name	From Approx. Sta.	To Approx. Sta.	Depth (mm)
Ellesmere Road	0+55	0+95	45mm from top – 90mm from top
Ellesmere Road South Side	1+17	2+80	40mm from top – 90mm from top
Ellesmere Road	3+74	7+43	40mm from top – 90mm from top
Ellesmere Road	8+27	11+80	45mm from top – 90mm from top
Ellesmere Road South Side	8+53	9+28	0 – 90mm from top

*Refer to the Asbestos Removal Plan for detailed areas.

*Refer to the Geotechnical Report for asphalt containing asbestos below 90mm from the existing surface.

The milling process shall be conducted at different depths as follows:

- For areas where asbestos is contained in 40mm from top to 90mm from top, mill to 40mm without asbestos and mill up to 60mm with asbestos.
- For areas where asbestos is contained in 45mm from top to 90mm from top, mill to 45mm without asbestos and mill up to 55mm with asbestos.

- For areas where asbestos is contained in 0mm from top to 90mm from to, mill up to 100mm with asbestos.

Immediately upon notification of award of the Contract and not more than 5 Working Days following the date of award notification, the Contractor shall prepare and submit to the City a detailed removal and disposal plan. The plan is to show the detailed procedure, sequence, protection of site personnel and timing of the work. Work cannot begin without an approved plan.

The Contractor shall remove and dispose the asbestos-containing asphalt pavement to the specified depth within the roadway. This milling operation shall not apply to the non-asbestos-containing streets in the Contract.

This special provision is applicable to the pavement adjacent to the curb and gutter and roadway structures on the streets containing asbestos, also applicable to top asphalt containing asbestos removal with milling operation and asbestos removal in the pavement structure within the trench limits.

The Contractor shall wet mill the asbestos containing asphalt pavement. Prior to removal of curb and asphalt containing asbestos, the Contractor shall wet mill with an edge grinder at a depth as specified in the table and a width of not more than 500 mm from the curb and around the perimeter of catch basins, manholes and any other roadway structures. Area air quality monitoring shall be done by the Contract Administrator if required.

The standard asphalt pavement removal and asbestos-containing pavement removal shall be completed in separate and distinct operations to ensure proper disposal of the waste material. A typical scenario would be to remove all standard surface asphalt first; then remove the surface and base containing asbestos; followed by removal of the standard base asphalt.

The Ministry of Labour Operational Approach

The Contractor shall comply with O. Reg. 278/05 "Designated Substance – Asbestos on Construction Projects and in Buildings and Repair Operations". As per the latest Ministry of Labour operational approach, measures and procedures, as outlined in Reg. 278/05, are to be followed in the performance of the work, as follows:

- Measures and procedures for Type 1 operations may be applied for operations carried out with power tools if the power tools are attached to dust-collecting devices equipped with HEPA filters or if the asbestos-containing asphalt is wetted to control the spread of dust or fibres.

- For non-classified operations, such as scarifying, milling and so on measures and procedures for Type 1 operations may be applied if the equipment is attached to dust-collecting devices equipped with HEPA filters or if the asbestos-containing asphalt is wetted to control the spread of dust or fibres.

- Measures and procedures for Type 2 operations may be applied for operations carried out with power tools not attached to dust-collecting devices equipped with HEPA filters and where the asbestos-containing asphalt is not wetted to control the spread of dust or fibres.

- For non-classified operations whereby the asbestos-containing material is not wetted to control the spread of dust or fibres and the equipment is not attached to dust-collecting devices equipped with HEPA filters, measures and procedures for Type 2 operations will continue to apply.

Measures and Procedures for Type 1 Operations

The Contractor shall carry out the asbestos-containing pavement removal operations in such a way that the measures and procedures for Type 1 operations can be applied.

In the event that a circumstance arises in which the Contractor cannot control dust (through either the attachment of HEPA-filtered dust collecting devices to the equipment or wetting), the Contract Administrator shall be notified and Type 2 measures and procedures shall be followed during the performance of the Work.

The Contractor must provide written notice of measures or procedures to be followed when performing the Work to the Contractor's joint health and safety committee/health and safety representative.

1. Health and Safety Training

Workers on the project must be trained in

- the hazards of asbestos exposure
- the use, care and disposal of protective equipment and clothing to be used and worn when doing the work
- personal hygiene to be observed when doing the work
- the measures and procedures prescribed by the regulation.

A letter shall be supplied to the Contract Administrator 7 Days prior to the start of asbestos removals outlining the compliance with the health and safety training.

2. Respirators

If the Contractor's worker request a respirator, the Contractor shall provide the workers with respirators according to Section 14, Paragraph 12 of O. Reg. 278/05. The respirators shall be as described in Section 13 and Table 2 in O. Reg. 278/05.

The workers who are using respirators shall follow the instructions described in Section 13, O. Reg. 278/05.

To address heat stress during hot weather, the Contractor must develop a hot weather plan and ensure that the plan is followed.

3. Protective Clothing

If the Contractor's worker requests protective clothing, the Contractor shall provide the workers with protective clothing according to Section 14, Paragraph 13 of O. Reg. 278/05. The protective clothing shall be as described in Section 15, Paragraph 12 of O. Reg. 278/05.

The workers who are using protective clothing shall follow the instructions provided in Section 14, Paragraph 14 of O. Reg. 278/05.

To address heat stress during hot weather, the Contractor must develop a hot weather plan and ensure that the plan is followed.

4. Eating and Drinking Prohibition

The Contractor shall advise their employees of the prohibition against eating, drinking, chewing or smoking in the work area.

5. Dust Control

The spread of dust from the work area will be prevented by the following dust suppressant control measures

- wetting down the work area prior to the start of operations.
- continued wetting throughout the duration of the operation by means of the equipment’s own wetting-down mechanism, in the case of the milling machine, and an available water truck.

- frequently and at regular intervals during the doing of the work and immediately on completion of the work, dust and waste shall be cleaned up and removed using wet sweeping, and placed in a container for asbestos waste.

Under no circumstances shall compressed air be allowed for any dust cleanup.

The Contractor shall prevent slurry from entering the sewers by placing geotextile or similar filters into affected catch basins. At the end of work, the filters shall be removed from the catch basins and deposited in containers for asbestos waste.

The Contractor shall submit a plan for site housekeeping that ensures efficient removal of the dust from the site and prevention of dust spreading into the environment.

6. Facilities for Washing

The Contractor will be required to have facilities on site for the washing of hands and face. All workers will be advised to use these facilities when leaving the work area.

7. Containers for Dust and Waste

Dust and waste shall be deposited immediately in a truck covered with a tarpaulin. The truck load shall be identified as asbestos waste, as required by R.R.O. 1990, Reg. 347.

Transportation and disposal of asbestos waste off site shall be according to item 8, herein. All costs associated with disposal are included in the Contract Price of milling asphalt items in this Contract.

8. Transportation and Disposal of Asbestos Waste

Blending of asbestos-containing and non-asbestos-containing asphalt waste in a single shipment for transportation and disposal is not allowed.

The transportation and disposal of asbestos waste under shall be managed in according to O. Reg. 347/90 “General – Waste Management” and the Transportation of Dangerous Goods Act.

All asbestos waste shall be disposed of at a site licensed for the acceptance and disposal of asbestos waste. The Contractor shall provide the City with the name and address of the waste disposal site at the pre-construction meeting.

Measurement and Payment

Measurement of the asbestos-containing asphalt material will be by area in square metres (m^2).

Basis of Payment

Payment at the Contract Price shall be full compensation for all labour, Equipment and Materials to do the Work.

The Contractor shall submit to the Contract Administrator all the weight tickets issued by the disposal site for the disposal of the asbestos wastes. Only the original weight ticket for each truck load will be accepted by the City. The Contractor shall be responsible for ensuring that the weight ticket for each load is handed to the Contract Administrator or City inspector on the same Working Day or before noon the following Working Day. The City will not compensate weight tickets not submitted at the proper time, or are submitted in groups after the milling and disposal operation.

The City reserves the right to randomly verify the quantity of asbestos waste transported to the disposal site. The Contractor shall permit the City to make random checks of the gross mass and the tare mass of trucks hauling the asbestos waste by requiring them to be driven over an independent scale. No separate payment shall be made for any delays or costs attributable to such verification of loads.

39. Concrete Road Base Repairs – SP#13

Non-Standard Special Provision — January 2019

After milling the road, the Contract Administrator and the Contractor shall walk through the site and determine any deficient areas or utility cuts that require reparations, adjacent to or within the construction limits. When instructed, the Contractor shall make all necessary reparations to the satisfaction of the Contract Administrator.

The cost of utility cut or full depth repair work (for any area, regardless of the size) on composite pavement shall refer to standard drawing T-509.010-1, T-508.010-1, Specification TS 2.10, TS 3.20 / TS 3.45 / TS 4.60/TS 1350/TS 3.40

Utility Cut and Road Restoration and additional notes below:

1. Saw cut along the limits of removal area (as directed by the Contract Administrator);
2. Remove full depth pavement (concrete / asphalt / u-fill) (full depth – up to 400 mm thickness);

3. If directed, sub-excavate an additional 150 mm, inspect the underlying material to ensure that it is suitable for pavement base/subbase and apply a reinforced geotextile (if necessary based on the inspection)
4. Supply, place and compact 150 mm thick, Granular A; (Payment for Granular A shall be as per the item "Granular A")
5. Construct high early strength concrete road base (32MPa at 7 Day) 250 mm or matching existing concrete road base, whichever is greater, including dowels as required. If the lane needs to be open within 4 days, 24 Hour concrete shall be used.
6. Supply and place rubberized asphalt joint along the edges of the concrete road base prior to placing base asphalt.
7. Dispose excess material off site.

Any full depth pavement repair work carried out shall be completed on the same day; no open excavations will be allowed after working hours. The Contractor shall carefully schedule the amount of repair work to be done within a day in order to meet this requirement. Contractor shall refer to the geotechnical report for the depth of removal, and the existing asphalt and concrete thickness, as the thickness of the asphalt and concrete varies from location to location. The repair area may be paved to the surface with base course asphalt temporarily if the adjacent road work includes the milling and replacement of the surface course asphalt.

Measurement for Payment

Measurement of composite pavement repair shall be by the surface area of the concrete repair or utility cut in square metres (m2). The additional area of the asphalt step joint is considered incidental to the work.

Basis for Payment

Payment at the Contract Price for the above tender item(s) shall be full compensation for all labour, Equipment and Material to do the Work, including the use of high early strength concrete.

Payment shall be based up to the depth shown in the appropriate tender item, except for depths in excess of the tender item with the largest depth which shall be prorated based on the tender item with the largest depth.

40. Flexible Pavement Repair – SP#14
Flexible Pavement Repair with Asbestos

This special refers to full depth pavement repair of flexible pavement in non-asbestos containing asphalt and asphalt containing asbestos. Both operations (asbestos and non-asbestos) shall not to be completed at the same time.

After milling the road, the Contract Administrator and the Contractor shall walk through the site and determine any deficient areas or utility cuts that require reparations, adjacent to or within the construction limits. When instructed, the Contractor shall make all necessary reparations to the satisfaction of the Contract Administrator. The repair must be in accordance to standard drawing T-509.010 (2 of 2) – Scheme A, Specification TS 4.60, except that HL-1 shall be replaced with the specified surface course asphalt for the area and HL-8 shall be replaced with the specified binder course asphalt for the area. The asphalt shall be restored to match the existing surface at the time of the repair.

Any full depth asphalt repair work shall be completed on the same day; no open excavations will be allowed after working hours. The Contractor shall carefully schedule the amount of repair work to be done within a day in order to meet this requirement. Contractor shall refer to the geotechnical report for the depth of removal as the thickness of the pavement material varies from location to location.

Measurement for Payment

Measurement will be the area of the full depth pavement repaired in square metres (m2). The additional area of the step joint is considered incidental to the work.

Basis of Payment

Payment at the Contract Price for the above tender item(s) shall be full compensation for all labour, Equipment and Material to do the work.

Payment shall be based up to the depth shown in the appropriate tender item, except for depths in excess of the tender item with the largest depth which shall be prorated based on the tender item with the largest depth.

41. High Early Strength Concrete – SP#15

Non-Standard Special Provision	January 2019

The supplemental cost items for high early concrete only apply if the tender documents do not already include the requirements for high early concrete in the items. Only one supplemental cost item will apply for each appropriate item.

Measurement for Payment

Measurement will be the volume of concrete placed in cubic metres (m3) and shall be calculated using the surface area of the component in square metres (m2) multiplied by the theoretic or actual thickness in metres (m), whichever is less. Concrete tickets shall not be used for measurement purposes.

42. Construction Specification for Pavement Marking – SP#16

Non-Standard Special Provision January 2019

Amendment to OPSS 710, November 2010

OPSS 710.01 CONSTRUCTION

OPSS 710.07.01 General

Subsection 710.07.01 of OPSS 710 is amended by the deleting the fourth paragraph in its entirety and replacing it with the following:

Where the pavement marking scheme is not shown in the Contract Drawings, the Contractor shall supply and place all pavement markings to the same size and location as the existing pavement markings. The Contractor shall be responsible to make appropriate sketches prior to construction so that the existing pavement markings can be fully reinstated.

Pavement marking shall be completed prior to opening roadway to traffic within 24 hours after paving operation finished.

OPSS 710.07.02 Surface Preparation

Subsection 710.07.02 of OPSS 710 is amended by the addition of the following paragraph:

The Contractor shall provide a mechanical sweeper to clean the road surface prior to the application of pavement markings. Pavement marking shall only be performed by an experienced and qualified Subcontractor. The work shall also include all required layout in conformance with the *Manual of Uniform Traffic Control Devices for Ontario*.

OPSS 710.07.03 Pavement Marking Obliterating

Subsection 710.07.03 of OPSS 710 is amended by the deleting the entire paragraph and replacing it with the following:

Pavement marking shall be obliterated or removed using an approved water blasting method only.

OPSS 710.07.05 Temporary Pavement Marking

Subsection 710.07.05 of OPSS 710 is amended by the addition of the following paragraph:

Temporary markings shall include the removal of the markings to facilitate traffic control. Temporary markings that are to be covered with an asphalt layer do not require removal. Temporary markings shall be restored within 72 hours of detecting any failure of pieces to remain according to specifications.

OPSS 710.07.06 Short Term Pavement Marking

Subsection 710.07.06 of OPSS 710 is deleted in its entirety and replaced with the following:

Short term pavement marking is required when an existing paved roadway, which is not to be reconstructed or resurfaced, requires pavement markings in different alignments and/or colours than the existing pavement marking.

As part of the work of pavement marking, the Contractor shall apply short term pavement markings for all affected markings, including centreline, lane lines, crosswalks and stop bars.

Short term pavement markings shall be applied according to MUTCD and as amended by Table 1 for Type C markings. The length of time before permanent markings are placed does not apply to this specification. The pavement markings shall stay in place as required. The Contractor shall inspect all short term pavement marking at the beginning and end of each week. The Contractor shall replace any failed pieces immediately.

Short term pavement markings shall not conflict with existing pavement markings. If required, the Contractor shall remove and replace the existing pavement markings as required.

Short term pavement markings placed on existing or on the final surface course shall be of the removable type.

OPSS 710.07.08 Selection of Materials

Subsection 710.07.08 of OPSS 710 is deleted in its entirety and replaced with the following:

All permanent pavement markings shall be field reacted polymeric pavement marking material according to OPSS 1714.

All temporary pavement markings shall be organic solvent based traffic paint according to OPSS 1712.

All short term pavement markings shall be temporary preformed plastic pavement marking tape according to OPSS 1715.

OPSS 710.07.09.02 Organic Solvent Based Traffic Paint

Clause 710.07.09.02 of OPSS 710 is amended by the addition of the following paragraphs:

A "centre liner" shall be used for the application of all temporary pavement markings to demarcate centre and through lanes. Hand paint machines may be used for all transverse pavement markings and symbols.

All temporary pavement markings, required for less than a three month duration, shall require only a single application. Temporary pavement markings required for longer periods shall require two separate applications.

OPSS 710.07.09.05 Field Reacted Polymeric Pavement Marking Materials

Clause 710.07.09.05 of OPSS 710 is amended by the addition of the following paragraph to the end of the first paragraph:

Markings shall not be applied to any joints between the asphalt and any appurtenance. Markings shall not be applied to the surface of any utility frame and covers.

Clause 710.07.09.05 of OPSS 710 is amended by the addition of the following paragraph to the end of the last paragraph:

The Contractor shall follow the manufacturer's guidelines for application on non-bituminous surfaces, that is to say concrete surface. Concrete surfaces require a primer/sealer to ensure adequate bonding of the permanent pavement marking.

OPSS 710.10 BASIS FOR PAYMENT

OPSS 710.10.01 Pavement Marking - Item
Pavement Marking Symbols - Item
Pavement Marking, Durable - Item
Pavement Marking Symbols, Durable - Item
Pavement Marking Temporary - Item
Pavement Marking Symbols, Temporary - Item

Contractor is liable for the cost of damages and repairs to these conduits as a result of damage by the Contractor's operations.

The Contractor shall satisfy themselves as to the actual location and depth of these utilities prior to excavation.

Basis for Payment

All costs associated with this work shall be considered incidental to all related items of Work, including all coordination, test pits, permits and application fees. No separate payment shall be made.

44. Working Around Enbridge Vital Gas Mains – SP#18

Non-Standard Special Provision	January 2019

The Contractor shall note that there are large gas mains within the limits of construction. A 300mm gas main running east west on Ellesmere Road is Enbridge Gas Distribution (EGD) asset.

The Contractor shall exercise utmost care to protect existing underground mains and vital mains during excavations and construction works in the vicinity of these utilities.

The contractor is responsible for ensuring appropriate measures are taken to protect and not to disturb the mains and vital mains. The contractor shall ensure that construction operations are performed without interfering with the continued safe operations of these existing underground mains and vital mains. The Contractor is liable for the cost of damages and repairs to these mains and vital mains as a result of damage by the Contractor's operations.

The Contractor shall locate all of these mains and vital mains and mark them on the ground. The Contractor shall contact the utilities and provide them advance notices as indicated below, to enable inspectors from these utilities be present on site during construction works:

- Enbridge Gas Distribution (EGD) – The Contractor shall contact Damage Prevention Representative at telephone 1-866-922-3622 by providing minimum 5 working days' notice prior to any construction work within 3m from either side of the Vital Main. The Contractor shall also provide this notice to Ontario One Call at telephone 1-800-400-2255.

The Contractor shall ensure the minimum vertical clearances from the vital mains are met, as specified in the crossing agreements. Test pits will be required for each vital main at each location requiring Ground Disturbance as per the National Energy Board (NEB) "Pipeline Damage Prevention". Pipelines that are parallel to the work may require additional test pits at regular intervals as determined by Enbridge "Third Party Requirement in the Vicinity of Natural Gas Facilities" or as required by the pipeline company to verify the location of the pipelines.

The Contractor shall conform to all existing legislation and guidelines regarding excavation or ground disturbance and damage prevention of all pipelines and vital mains. We have attached the following publications acknowledging that this is not a complete set of the required legislation and guidelines:

- Enbridge "Contractor Safety Manual (Canada)" ;
- Enbridge "Third Party Requirement in the Vicinity of Natural Gas Facilities"; and
- NEB "Excavation and Construction Near Pipeline".
- NEB "Pipeline Damage Prevention – Ground Disturbance, Construction and Vehicle Crossings ".
- Technical Standards & Safety Authority "Guideline for Excavation in the Vicitnity of Utility Lines"
- For Horizontal Directional Drilling (HDD) or Directional Boring, an additional daylight hole(s) is required to visually verify the drill head's location (including depth) relative to the measurement of the tracking equipment.
- The Contractor shall determine through the utility contacts, their requirements and if fees are applicable, as potential vital main inspection may be required for when the excavation is occuring.

Basis for Payment

All costs associated with this work shall be considered incidental to all related items of Work, including all coordination, test pits, permits and application fees. No separate payment shall be made.

The cost charged by the utility company for providing their inspectors shall be covered under a separate tender item. The Contractor shall alter his methods to minimize the time required by the utility company inspectors. The Contractor will be responsible for all utility company inspection costs if it is deemed by the Contract Administrator that the Contractor to not take appropriate measures to minimize the time requirement or as a result of penalties imposed.

45. Tree Growing Medium, 800mm Thick – SP#19

Non-Standard Special Provision March 2020

Tree Growing Medium item in the Pricing Form shall be installed in soil replacement areas identified on Contract Drawings 20-02980-003 and 20-02980-004. Type 1 - Standard Mix shall be used as the tree growing medium. Specifications of the tree growing medium shall in accordance with TS 2.10 and TS 5.10 with the following additional requirements:

- Existing soil removal and placement of the tree growing medium (Type 1 – Standard Mix) shall be as per the Detail C on Contract Drawing 20-02980-007.
- The bottom and sides of the excavation shall be scarified so that there is no smooth surface.
- The tree growing medium shall be installed in 300mm layers and properly compacted in between as specified in TS 5.10.
- Nursery sod shall be installed on top of the tree growing medium.

Measurement for Payment

Measurement of Tree Growing Medium shall be by surface area in square metres (m^2)

Basis for Payment

Basis of payment shall be as stipulated in TS 5.10. "Basis of Payment" shall be amended to include "Tree Growing Medium, 800mm Thick" as specified above and nursery sod.

46. UniGranite Unit Paver – with Concrete Base – SP#20

Non-Standard Special Provision March 2020

UniGranite Unit Pavers shall be in accordance with TS 3.80, except that:

- UniGranite Unit pavers from Unilock shall replace the concrete unit pavers; and
- Pavers shall be dark charcoal meeting the dimensions of 10 cm x 10 cm x 7 cm depth.

UniGranite Unit Pavers are intended to be installed as a buffer strip between the proposed / existing sidewalk and proposed multi-use trail. The paver strip shall be constructed as per Detail A and B, as shown on Contract Drawing 20-02980-007.

The surface of the UniGranite pavers shall be flush with the adjacent sidewalk and multi-use trail.

47. Test Pits to Expose and Verify Existing Utilities – SP#21

Non-Standard Special Provision March 2020

The Contractor shall arrange for non-mechanical type excavations, such as Hydrovac or hand digging to undertake the exposure of the existing utilities or structures within the roadway or boulevard and as required by utility companies.

A plan of the proposed locations of all test pits is to be supplied by the Contractor to the Contract Administrator. Prior to proceeding with the proposed test pits approval from the Contract Administrator is required.

The test pits to expose and verify utilities shall include the following:

all traffic control;

excavation, sheathing, shoring, trench dewatering;

the removal and disposal off site of asphalt pavement, concrete road base, brick gutter, concrete or asphalt curb, curb and gutter, monolithic curb and sidewalk, sidewalk, crosswalks, walkways, flag stone, boulevards, driveways and entrances (all thicknesses);

the removal and disposal off site of asphalt pavement containing asbestos in accordance with Ontario Regulation 278/05, Type I;

remove and salvage existing unit pavers and interlock stone;

protecting and supporting of adjacent components that shall remain during and after the work including existing structures, sewers, laterals and watermains, services and utilities, light poles, hydro and traffic signal poles, hand wells, maintenance holes, catch basins, retaining walls, curbs, road base, driveways, sidewalks, crosswalks, walkways, valve boxes, signs and posts, fences, bollards, gardens, decorative stones, garden edging and vegetation, residential sprinkler systems and lighting systems. Any damage to existing components shall be properly repaired at no extra cost to the City;

disposal of excavated material off site;

the supply, placing and compacting of unshrinkable fill according to TS 13.10 under hard surfaces such as asphalt, concrete, interlock stone and Granular B Type II or select native material at other locations for backfilling;

the installation and maintenance of permanent restoration for concrete and asphalt sidewalks, curb and roadway;

permanent restoration of all existing landscaping, sodding, fences, river stone, gardens, decorative rocks, vegetation, interlock stone or lock stone, unit pavers, concrete and asphalt walkways or driveways (all types), sodding to pre-construction conditions or better.

A minimum of 300 mm of compacted Granular A (free of RAP; reclaimed asphalt pavement) material shall be placed around any water services before unshrinkable fill is placed.

For gas mains, a minimum of 150 mm of sand padding must be placed over the pipe for protection. The gas main and valve assemblies must be sand padded before placement of the unshrinkable fill. The Contractor must ensure that placement of the unshrinkable fill does not displace the sand padding or directly contact the pipeline.

Test pits agreed to or requested by the Contract Administrator shall be at the Contract Price for test pits as specified in the Contract Documents. Test pits required by the Contractor to confirm locates are not payable.

The following test pits shall be undertaken prior to construction to verify utilities:

Location	Test Pits Required
Ellesmere Road North boulevard at approx. Sta. 3+55	One (1) Test Pit up to 1.2m deep to verify Bell conduit
Ellesmere Road North curb lane at approx. Sta. 4+07	One (1) Test Pit up to 2.0m deep to verify Enbridge gas main and CB Lead
Ellesmere Road North boulevard at approx. Sta. 4+50	Two (2) Test Pits up to 1.2m deep to verify Bell conduits
Ellesmere Road North boulevard at approx. Sta. 7+40	One (1) Test Pit up to 1.2m deep to verify Bell conduit

*Refer to the contract drawings for detailed locations

Measurement for Payment

For measurement purposes, a count shall be made of the number of authorized test pits completed.

Basis of Payment

Payment at the Contract Price shall be full compensation for all labour, Equipment and Material to do the Work.

Where the removal, disposal, backfill and restoration for the test pits are included in other tender items, the payment for the work shall be under the appropriate tender items.

48. Tree Protection – SP#22

Non-Standard Special Provision March 2020

Tree Protection Plan

A Tree Protection Plan shall be submitted and approved prior to construction that will describe how the Contractor shall protect trees from damage. An arborist may need to be consulted to develop the Tree Protection Plan. It should be noted that in some situations where Work will be undertaken within a Tree Protection Zone, erecting tree protection fencing may not be feasible. Maximum depth of excavation and other methods to minimize the impact to the tree roots shall be addressed in the Tree Protection Plan. For more information on the Tree Protection Policy and Specifications for Construction near Trees (July 2016) go to www.toronto.ca/data/parks/pdf/trees/tree-protection-specs.pdf .

No work shall commence within the dripline of any tree until a Tree Protection Plan has been reviewed and approved by the Contract Administrator.

The following tree protection requirements in the north boulevard of Ellesmere Road have been identified by the City:

Approx. Tree Location (Station)	Tree Protection Zone (TPZ) Radius (m)	Encroachment (m)
0+72	1.2	
0+82	1.2	
0+93	1.2	
1+20 (tree to be injured)	1.2	0.2
1+30 (tree to be injured)	1.8	0.7
1+49 (tree to be injured)	1.8	0.8
1+58 (tree to be injured)	1.8	1
1+67 (tree to be injured)	1.8	1
2+04	1.2	
2+20	1.8	
2+62	1.2	
2+86	1.2	
3+00	1.2	
3+10	1.2	

3+22	1.2	
3+36 (tree to be injured)	1.8	0.5

1. TPZs must be built to the size of the area outlined in the table above;
2. For the trees indicated to be injured, the TPZs are to be built at 0.2m to the edge of the proposed multi-use trail;
3. Work done within the encroachment area of trees being injured is to be completed by hand, roots cleanly pruned before larger machinery is to be brought through this area.

Basis of Payment

All costs associated with developing and implementing a Tree Protection Plan is considered incidental to all Work. No separate payment shall be made.

Tree Protection

In some situations where work will be undertaken within a Tree Protection Zone or large areas with little work is required in the vicinity of trees, adhering to the *Tree Protection Policy and Specifications for Construction Near Trees* may not be feasible. In these situations the following tree protection guidelines shall be followed:

1. The Contractor shall conduct operations in such a manner as to not cause damage to trees, above or below grade.

2. If required, any pruning of roots or crown must be done using best arboricultural practices.

3. All arboricultural work, including pruning of roots or limbs greater that 5 cm in diameter, shall be completed by an arborist who has one of the following qualifications: qualified by the Ontario Training and Adjustment Board Apprenticeship and Client Services Branch, is a certified arborist qualified by the International Society of Arboriculture, is a consulting arborist registered with the American Society of Consulting Arborists, is a registered professional forester or a person with other similar qualifications as approved by the General Manager, Parks, Forestry & Recreation.

4. Roots requiring pruning by an arborist must first be exposed using pneumatic (air) excavation, by hand digging or by using a low pressure hydraulic (water) excavation. The water pressure for hydraulic excavation must be low enough that root bark is not damaged or removed. This will allow a proper pruning cut and minimize tearing of the roots.

5. No storage of equipment, materials, excavated soil, construction waste or debris is permitted within the Tree Protection Zone.

6. No compaction of soil is permitted within the Tree Protection Zone. If equipment is required to travel within the Tree Protection Zone, horizontal hoarding must be utilized to prevent soil compaction.

7. Roots that have been exposed and pruned by an arborist must be watered daily until backfilled.

8. All roots and branches pruned by an arborist shall be removed from public view immediately and must be removed from the project site by the end of each work day.

9. Permits will be required for privately-owned trees, protected under the Toronto Municipal Code, Chapter 813, Trees, Article III, Private Tree Protection and Chapter 658, Ravine and Natural Feature Protection bylaws, where work is proposed within the Tree Protection Zone of the private trees in question.

10. In the event a tree is damaged in any way, including mechanical injury to the main stem, tearing of limbs or roots with a diameter of 5 cm or greater, the damage must be reported to the Contract Administrator immediately. All work in the area must cease until the Contractor has the damage assessed by an arborist and any remedial work recommended by the arborist is completed. The arborist shall submit a report of the incident and actions taken to the Contract Administrator.

11. If proper tree protection practices are not in place and tree damage has occurred as a result of the Contractor's negligence, Urban Forestry staff may issue stop work orders. Orders to comply may include the need to erect hording around trees in accordance to the *Tree Protection Policy and Specifications for Construction Near Trees* and require the submission of contravention inspection fees.

Basis of Payment

All costs associated with this work, including the cost of an approved arborist, is considered incidental to all Work. No separate payment shall be made.

49. Concrete Sidewalk – SP#23

Non-Standard Special Provision — May 2020

At driveways to be replaced in Phase 2, 2" of temporary asphalt patching beside the sidewalks will be required if the sidewalk is reconstructed. This will be at no additional cost.

Amendment to TS 3.70 September 2017

TS 3.70.09 MEASUREMENT FOR PAYMENT

TS 3.70.09.01 Concrete Sidewalk Concrete Raised Median

Subsection 3.70.09.01 of TS 3.70 is amended by the addition of the following:

Measurement of concrete sidewalk and raised median placed shall be by surface area in square metres (m^2), measured from the back of curb (typically 200 mm from face of curb) or face of structure. No deductions will be made for maintenance holes and other appurtenances.

TS 3.70.10 BASIS OF PAYMENT

TS 3.70.10.01 Concrete Sidewalk – Item

Subsection 3.70.10.01 is amended by deleting the first paragraph in its entirety and replacing with the following:

Payment at the Contract Price for the above tender item shall be full compensation for all labour, Equipment and Material to do the Work. Payment shall include all excavation, the supply, placing, levelling and compacting of the granular base, the supply, placing and removal of the formwork, the supply, placing, consolidating and finishing of the concrete and the curing and protection of the concrete sidewalk.

50. Concrete Curb and Gutter – SP#24

Non-Standard Special Provision May 2020

Concrete curb and gutter shall meet the requirements of TS 2.10 / TS 3.17 / TS 3.50 / TS 1350 and T-600.05-1, except that the height of the new curb shall match the existing curb height, which is generally 130mm.

51. Signal And Electrical Works General Information For Tenderers – SP#25

Non-Standard Special Provision May 2020

a) All traffic signal work shall be completed by approved Contractors listed in Part 4 - Form C - Experience and Qualifications Requirements

b) The figures shown for the estimated quantity are for purposes of tender assessment only and shall not be taken as a guarantee of the actual work to be performed under this contract which may, as circumstances require, be either more or less than the indicated amounts.

c) The estimated quantities shown in the Form of Tender are subject to increase or decrease.

d) All Unit Tender Prices on the Request for Quotation shall be full compensation for all labour, Equipment and Material to do the Work including but not limited to the following:

- all labour, vehicles, equipment and tools to complete the work
- all materials, unless otherwise specified
- supervision
- attendance on site by Contractor staff during final inspection
- work completed to City of Toronto Transportation Service Traffic Control Device Specifications.
- travel time to and from the site
- work zone safety set-up and equipment
- assembly, preparation, installation and activation of components
- site meetings as required
- meeting with local hydro to co-ordinate hook-up as required
- all testing as required
- restoration of the site to original or better condition
- returning of salvaged material to Toronto Transportation
- removal and disposal of damaged/obsolete equipment
- removal and disposal of debris
- procurement of Electrical Safety Authority permit, City of Toronto utility cuts permit and any necessary permits as required including the permanent restoration costs of road crossings.
- restoration costs of roadways, sidewalks and boulevards
- all specified documentation
- obtaining any necessary utility clearances
- obtaining of any approvals to enter local hydro plant facilities
- applicable Provincial Sales Tax
- all Overhead costs including the following:
    - all tools required to facilitate the works
    - Paid Duty Police, as required.
    - contract administration
    - bonding charges and insurance fees
    - arrangement for permits
    - all management, accounting and data entry
    - office space, store-keeping and housekeeping
    - supply and service charges for telephones, fax machines and mobile two-way radio and cellular phone equipment
- All Payroll costs including the following:
    - Employer's Health Tax
    - Workers Compensation

- Unemployment Insurance
- Canada Pension Plan
- and any other Payroll Burdens that apply to the quoted labour rates

No extra charges of any kind shall be permitted or accepted.

52. Signal And Electrical Works Special Specifications – SP#26

Non-Standard Special Provision — May 2020

<u>TRAFFIC CONTROL SIGNALS – PERMANENT & TEMPORARY</u>

ESP1 ELECTRICAL WORK GENERAL FOR TRAFFIC SIGNALS

- Per TTS 801.
- TTD 801.100 shall apply to all work as indicated on the drawings.
- Reprogram existing timer to new timing card program provided by the City.
- Contractor to inspect existing conduits, where new wiring is required. The inspection shall be in the presence of the City's electrical inspector.
- The work shall include coordination with ESA and Toronto Hydro Energy Services, utility stake-outs, permits, fees and certificates, layout of electrical work, quality control and quantity assurance.
- Paid duty Police Officer, if required.
- Price to include the 24/7 maintenance of existing and temporary traffic signals during construction. Traffic signals must be operating at all times throughout construction except for switchovers during stages.
- Payment for lump sum contract price for the work specified herein shall be full compensation for all labour, equipment and material required to do the work.
- Price to include any temporary cables, junction boxes etc. to maintain existing traffic signal operation during construction.

ESP2 CONCRETE “TRAFFIC” HANDWELL COMPLETE ASSEMBLY, INCLUDING FRAME AND COVER AND GROUNDING LUG – 460mm INSIDE DIAMETER

- Per TS 802.
- Dimensions of handwell to comply with TTD 802.006.
- Handwells can be cast-in-place or precast.
- Precast handwells only to be installed in boulevard areas with surface cover of sod or asphalt.
- Cast-in-place handwells to be installed in sidewalks or concrete covered boulevard areas.
- Unshrinkable fill is to be installed around precast handwells and, as required, around cast-in-place handwells.
- Lifting hooks on precast handwells to be removed after installation.

- Only approved suppliers of precast handwells are to be used as per OPSS Designated Sources list for OPSD 2112.02 (modified to TTD 802.006).
- Frames and lids are to be included in price.
- Where a ground rod is required at a handwell, price to include #6 bare ground wire.

ESP3 1 x 75mm DIAMETER PVC RIGID ELECTRICAL DUCT IN BOULEVARD (CONCRETE/PAVING STONE SURFACE) INCLUDING BOULEVARD SURFACE RESTORATION MATCHING EXISTING

- Per TS 803.
- This specification describes the requirement for the installation of rigid duct direct buried. Rigid duct shall be rigid PVC heavy wall type and shall meet the requirements of C.S.A Standard C22.2 No. 211.2.
- Duct to be Rigid PVC unless otherwise specified.
- Boulevard surface cover type applies to existing cover.
- Ends of all ducts shall be temporarily plugged or sealed until wiring is installed. See drawings for layout of wire and conduit runs.
- Fittings to be manufactured for use with conduit specified and have the same coating as conduit. Factory “ells” shall be used where 90 degree bends are required and knock-out fillers to be provided to prevent entry of foreign materials.
- Price to include duct.
- Price to include the cost of sidewalk restoration and boulevard repairs where applicable.
- Measurement for ducts will be made horizontally in metres, along the longitudinal axis of the duct and shall be from end to end of ducts, where the ducts do not terminated in a structure or, from center to center of pole, pole footings, electrical chambers and concrete pads.
- Payment shall be made as the unit price bid for this item. The unit price bid shall include all labour, equipment and materials required to do all work regardless of the sizes for ducts in any one run, including sand bedding, protection, compaction, cable protection tiles, locating of existing ducts and breaking into concrete structures and the removal of pavement, sidewalk, curb and curb and gutter.

ESP4 12 CONDUCTOR CABLE – TRAFFIC SIGNAL DISPLAY, IN DUCT
ESP4 12 CONDUCTOR CABLE – BICYCLE SIGNAL DISPLAY, IN DUCT
ESP4 7 CONDUCTOR CABLE – PEDESTRIAN SIGNAL DISPLAY, IN DUCT

- Per TS 804 and TS 810 as specified.
- Traffic signal and Bicycle signal cables shall be IMSA Spec 12 - 1/C # 14 AWG.
- Price shall include the cost of riser conduit and fittings, riser cable where applicable.
- Payment shall be made as the unit price bid for this item. The unit price bid shall include all labour, equipment and materials required to do all the work.

ESP5 2 CONDUCTOR #14 EXTRA LOW VOLTAGE - APS PUSHBUTTON

- Per TS 810 as specified.

- Traffic signal cables shall be IMSA Spec 19-1C.
- Price shall include the cost of riser conduit and fittings, riser cable where applicable.
- Contractor to remove existing loop cable and install new cable in same position as old loop.
- Payment shall be made as the unit price bid for this item. The unit price bid shall include all labour, equipment and materials required to do all the work.

ESP6 #6 AWG INSULATED GROUND CABLE – IN DUCT

- Per TS 813.
- Measurement of ground wire is horizontally in metres from ground rod to ground rod or centre of structure or pole.
- Price to include cost of riser ground wire where applicable.
- #6 ground wire to be RWU90 insulated stranded ground wire.
- Payment shall be made as the unit price bid for this item. The unit price bid shall include all labour, equipment and materials required to do all the work.

ESP7 GALVANIZED STEEL SIGNAL POLE – 6.1m (20ft), HEAVY DUTY

- Per TS 805.
- Galvanized Steel Traffic Signal poles to be supplied by Contractor per TTS 805.200
- Galvanized Steel Traffic Signal poles to be Valmont, Dynamental, or approved equivalent
- Payment shall be made as the unit price bid for this item. The unit price bid shall include all labour, equipment and materials required to do all the work.

ESP8 CONCRETE FOOTING FOR BASE MOUNTED POLE (600mm x 2500mm)

- Per TS 807, TTD 807.001.
- Concrete is to be vibrated to eliminate voids, honeycomb, and entrapped air.
- Anchorage assemblies included, Richmond type anchorage for pole bases.
- Provide an additional 50mm rigid duct sleeve at traffic/light combination pole base for lighting use.
- Price to include the cost of sidewalk restoration and boulevard repairs where applicable.
- Payment shall be made as the unit price bid for this item. The unit price bid shall include all labour, equipment and materials required to do all the work.

ESP9 TRAFFIC SIGNAL MAST ARMS, 3.7m (12ft)
TRAFFIC SIGNAL MAST ARMS, 4.6m (15ft)
TRAFFIC SIGNAL HEAD CUSHION HANGER
300mm DOUBLE ARM BRACKET (SET)
ASTRO BRACKET – BICYCLE SIGNALS

- Per TS 808.
- Price to include cost of double arms.

- Price to include the cost of riser cable.
- Price to include the cost of mounting all sign(s) complete with mounting hardware where applicable.
- Payment shall be made as the unit price bid for this item. The unit price bid shall include all labour, equipment and materials required to do all the work.

ESP10 LED HIGHWAY TRAFFIC SIGNAL HEAD COMPLETE WITH BACKBOARD LED COUNTDOWN DISPLAY PEDESTRIAN SIGNAL HEADS (2 SECTION) LED BICYCLE SIGNAL HEAD (BLACK) WITHOUT BACKBOARD

- Per TS 808
- Price to include cost of riser cable and riser conduit and fittings where required.
- Bicycle Signal heads shall be installed without a backboard. The bicycle signal head shall be black.
- Payment shall be made as the unit price bid for this item. The unit price bid shall include all labour, equipment and materials required to do all the work.

ESP11 AUDIBLE PEDESTRIAN APS PUSHBUTTON, CENTRAL CONTROL UNIT, INTERFACE CARD, AND ASSOCIATED WIRING

- Per TS 810 and TS 814
- Install Accessible Pedestrian Signal Unit – 2 Wire System (Polara).
- Price to include the installation of the pedestrian information sign to the audible unit:
  - The City of Toronto is using two types of signs: "Blue" and "Black and white".
  - Blue signs are for fixed time push button direction and no need to push the button. The walk time will come automatically. Black and white sign is for semi actuated intersections.
- The APS unit must be installed such that the surface of the pushbutton is aligned with the corresponding crosswalk.
- The height of the Accessible Pedestrian Pushbutton has to be 1.05m (Per 810.003)
- Payment shall be made as the unit price bid for this item. The unit price bid shall include all wire, labour, equipment and materials required to do all the work.

ESP12 OVERHEAD WAVETRONIX RADAR DETECTORS

- Radar detectors shall be supplied and installed at locations and mounting heights at the location(s) specified on the Contract Drawings. Radar detectors shall be orientated and configured for operation according to the manufacturer's specifications and the Contract Documents.
- Price to include installation, setup, programming of controller and activation of Non-Intrusive Detector that detects vehicle(s) waiting at the intersection via processing of detection images. The detector sensors may include, but shall not be restricted to these models:

Manufacturer	Wavetronix

Sensor Model	Wavetronix Matrix including Mounting Bracket, Junction Box, and 40' Cable.
Interface Model	Click 650 (CLK650), communicates directly with controller via SDLC.
Cable Required	20 awg 2 Twisted Pair with Drain Complete with Connectors

- Radar vehicle detector units shall have an adjustable detection pattern and angle. Units shall be complete with heavy duty mounting bracket suitable for pole mounting.
- The radar vehicle detection system distributed by Fortran Traffic Systems Limited is approved for use.
- Price to include installation of all components and supply of all cabling;
- Price to include assessment and investigation of each location;
- Price to include modifying hard copy of programming sheet, taking a picture and sending to project manager;
- Price to include cable lashing, clamps and cable supports;
- Price to include all remapping of timer;
- Price to include connecting SDLC cable to Wavetronix Click 650 interface;
- Enabling BIU 10 (if required) in the timer for Wavetronix Matrix installations;
- Price to include soldering to the detector rack and/or to the test switch door if necessary;
- Price to include the installation of approximately 200 m interconnect cable per intersection installed through underground duct system, or overhead messenger cable. Interconnect/homerun cable to be supplied by the Contractor;
- Price to include calibration and configuration set-up of the system upon installation, activation of the system and verification of operation;
- Price to include the cost of adjusting and aiming detection area;
- Price to include the cost of configuring the interface card/Click 650 and setting up detection zones;
- Price to include calibration of detection system and verification that the sensor(s) are working by verifying the operation with CSO desk;
- Price to include field analysis in regards to the new installation of detection systems per intersection including a wiring diagram to achieve the required detection operation;
- Price to include all labour, vehicles, equipment and tools required to complete work;
- Price to include the disconnection of loop detectors in the cabinet and removal of detector cards
-
- All cabling and equipment required to transmit, receive, and process radar detection signals shall be installed according to the manufacturer's specifications and the Contract Documents. Connection of the Smart Sensor power and communication cable too the shielded extra low voltage cable shall be completed in the pole handhole inn metal poles and in a junction box on wooden poles.

- The Operating Authorities Traffic Signal Maintenance Contractor shall test the radar overhead detector equipment for the desired range. The Contractor shall make any adjustments necessary, as specified by the Operating Authority. The testing and adjustments to the desired range shall be completed on the day the detection equipment is activated.
- Cables and connectors need to be as per the manufacturer's standards. No substitution of cables or connectors is permitted.
- Payment shall be made as the unit price bid for this item. The unit price bid shall include all labour, equipment and materials required to do all the work.

ESP13 MODIFICATIONS TO EXISTING TRAFFIC SIGNAL CONTROLLE AND CONTROLLER CABINET INC. PROGRAMMING

- This item is for the set-up and testing of, and all modifications to the traffic signal controller and cabinet including programming and inserting any new timing card as required by the contract.
- The set-up of the existing traffic signal controller shall be carried out by the Contractor in the presence of Contract Administrator.
- Contractor will have to supply and install a shelf above the controller to mount additional equipment i.e. pedestrian station, etc.
- This item also includes any modifications to the traffic signal controller cabinet, including wiring, necessary to implement proper operation of the traffic signal for all the new equipment and accessories.

Basis for Payment

- Payment shall be made at the applicable lump sum price for each traffic signal installed to set-up and modified for the controller and cabinet.
- Payment at the lump sum price shall be full compensation for all labour, equipment and material required to complete this work as specified in the Contract Documents.

ESP14 INSTALL / RELOCATE ROADWAY SIGNAGE

- This item is for the installation of all permanent roadway signing for the Contract as indicated on the electrical drawings.
- The Contractor shall carefully remove all regulatory, warning and information signing, including supports, which are affected by construction activities. These signs shall be removed from their supports and relocated to new locations identified on the Contract Drawings.
- The Contractor will manufacture any proposed new signs and any new signs required to replace the existing signs. The Contractor shall order the necessary signs based on the Contract Drawings.

References

- OPSS 510-Removals
- Ontario Traffic Manual (OTM) – Books 5 and 6
- MTO Sign Support Manual, Ministry of Transportation in Ontario, April 2007

- Occupational Health and Safety Act

Basis for Payment

- Payment for this item shall be made at the applicable lump sum price and shall be full compensation for all labour, materials and equipment required to complete the Work as specified in the Contract Documents. Progress payments shall be based on the estimated percentage of the total Work for this item completed to date. In no event shall the amount paid under this item exceed the total lump sum amount for this item.

ESP15 REMOVE AND SALVAGE / DISPOSE OF EXISTING TRAFFIC SIGNALS AND HARDWARE

- Per TS 815.
- The Contractor shall remove and salvage/dispose all existing and temporary traffic signal works including pole bases, handwells and any other associated concrete enclosures, with the exception of that intended for the permanent traffic signal systems as shown on the drawings.
- Existing overhead detectors including controller interface that are removed shall be returned to the City Traffic Signal Management.
- Payment for lump sum contract price for the work specified herein shall be full compensation for all labour, equipment and material required to do the work.

ESP16 REMOVE AND SALVAGE / DISPOSE OF EXISTING GUY

- The Contractor shall remove and salvage/dispose the existing GUY wires at the northwest corner and northeast corner of Ellesmere Road and Dormington Drive intersection.
- Payment shall be made as the unit price bid for this item. The unit price bid shall include all labour, equipment and materials required to do all the work.

this list of confined spaces. Without limiting the foregoing, the identification of confined spaces by the City was based on certain conditions that existed at the time the assessment was conducted, which conditions may be different or may change during the course of the performance of the Work under the Contract. The City assumes no liability whatsoever arising out of or in connection with this list of confined spaces or any reliance thereon.

b. The provision of this list of confined spaces in no way limits the Contractor's obligations as employer and, where applicable, as constructor under the OHSA, in particular those obligations with respect to confined spaces. The Contractor, at no additional cost to the City, shall be responsible for making its own assessment as to which spaces are confined spaces at the project site, including any new confined spaces that are created from time to time as construction progresses. Without limiting the foregoing, the Contractor shall not make any claims for delays or extra costs as a result of having to perform its obligations under the OHSA with respect to confined spaces.

c. The Contractor shall keep available for inspection at the project site every assessment, plan, co-ordination document, training record, entry permit, inspection record, and test record as required under the OHSA. Such documents shall be made available to designated City staff and consultants at the project site in the event that any City staff or consultants wish to enter any confined spaces at the project site for inspection and quality control purposes. The Contractor shall also provide to the Contract Administrator its own list of any confined spaces it has identified at the project site before the Work begins, and shall immediately notify the Contract Administrator in writing of any changes to this list from time to time during the course of the construction, and on completion of the project.

DS-4 Asbestos

a. Where the Work includes removal of asbestos, the Contractor shall:

   i. ensure, through appropriate air testing and such other measures as may be appropriate and necessary, that the Work site and adjacent areas not been contaminated with asbestos during the performance of the Work; and

   ii. prior to dismantling any barriers erected to contain asbestos and asbestos-containing materials, the Contractor shall provide written confirmation to the Consultant that, after conducting proper air testing and other due diligence measures, the area is safe in accordance with the requirements of the OHSA.

b. If, during the course of the Work, the Contractor or any of the subcontractors or suppliers engaged by the Contractor, disturb material that is believed to be asbestos containing material, separate and apart from asbestos abatement work forming part of the Contract, the Contractor shall act in strict compliance with the OHSA, including but not limited to the Asbestos Regulation, and without limiting the generality of the foregoing, shall:

i. Stop work and evacuate the area where the asbestos containing material is believed to have been disturbed and take all precautions or actions mandated by the OHSA and notify the City immediately;

ii. Notify the Contract Administrator via telephone, with written notification to follow as soon as possible; and

iii. Refrain from entering the work area for any reason whatsoever until safe to do so, in accordance with the requirements of the OHSA and, prior to re-entry, notify the Contract Administrator for approval to recommence Work.

DS-5 Workforce Development Plan

a. Where required in the Agreement Documents, the Contractor shall implement and document the Workforce Development Plan described in "Schedule F – Owner Policies, Procedures, by-Laws and Other Requirements" to the satisfaction of the City (in its sole discretion).

DS-6 Coordination and Meetings

a. The Contractor shall attend regular meetings with the City of Toronto and others, including but not limited to, Toronto Transit Commission, Bell Canada, Enbridge, Toronto Hydro, and business organizations as may be required by the Contract Administrator to co-ordinate services affected by the Contract and to monitor on-going administration and progress of the contract.

DS-7 Standard Specifications and Standard Drawings

a. The City's Standard Specifications and Standard Drawings that apply to the Work shall be those that can be found on-line at www.toronto.ca/ecs-standards as of the date the tender for the Work is issued.

b. Any other required work, for which no specifications are contained herein, shall conform to the City of Toronto Standard Construction Specifications and Drawings for Road Works, the City of Toronto Standard Construction Specifications and Drawings for Sewers and Watermains, the Ontario Provincial Standard Specifications and the Ontario Provincial Standard Drawings.

c. This Agreement may also refer to Ontario Provincial Standards (OPS) specifications and drawings. In such case, Bidders shall acquire the applicable specifications and drawings from OPS. Information about OPS can be found at www.ops.on.ca.

DS-8 Payroll Burden Rate for Work on a Time and Material Basis

a. Standard Rate (40%)

i. The Owner will pay the Contractor's Payroll Burden at a standard 40 per cent of the wages and salary portion of the Cost of Labour for change in the work in the Contract that is carried out on a Time and Material basis.

b. Option for Contractor’s Actual Payroll Burden Rate

i. Alternatively, the Owner will consider paying at the Contractor’s actual payroll burden rate. To be considered for this option, the Contractor MUST submit their actual payroll burden rate on the Owner’s prescribed Contractor’s Payroll Burden Form ("Form") prior to the commencement of any work on a Time and Material basis, preferably at the pre-construction meeting.

ii. The Form is available from the Contract Administrator upon request and it shall be completed, certified and signed by the Contractor’s external auditor. The Payroll Burden rate shall be calculated from the total expenditures of wages, salaries and benefits for all of the Contractor's employees paid during the previous 12 month calendar year (i.e. January 1st to December 31st). All permitted expenses in relation to labour costs are included on the prescribed Form.

iii. If accepted, the submitted Form shall be effective until January 31st of the following year and the payroll burden rate will apply to all Time and Material works carried out within the effective period of the Form. If the Contractor fails to submit a signed Form before the commencement of any work on a Time and Material basis, or if the submitted Form is not acceptable to the Owner, the Owner will apply the 40 per cent standard payroll burden rate for all works that are carried out on a Time and Material basis under this Contract until a Form is submitted by the Contractor and accepted by the Owner.

iv. During the Contract period, the Contractor must submit an updated Form by January 31st of a new calendar year. If accepted, the updated Form shall be effective until January 31st of the following year. If the Contractor failed to submit an updated Form or the submitted Form is not acceptable, the Owner will apply the standard 40 per cent payroll burden rate to all Time and Material works carried out under this Contract until an updated Form is submitted by the Contractor and accepted by the Owner.

v. The Owner reserves the right to terminate the application of the Contractor’s actual payroll burden rate and apply the standard 40 per cent payroll burden rate if the Form is found to be not accurately completed after its acceptance.

vi. Contractor’s labour rates used in the work based on a Time and Material basis are subject to verification by the City’s Fair Wage Office.

vii. All information in relation to Contractor’s Payroll Burden may be audited at the Owner’s discretion. The Contractor agrees to keep complete and accurate books, payrolls, accounts and employment records and make the records available for audit by the Owner upon request. The Owner reserves the right to recover any overpayment to the Contractor affected by the audit.

DS-9 Organization of Work and Work Restrictions

a. It is the Contractor's responsibility to implement all required measures (e.g. fences, enclosures, etc.) in order to strictly control the pedestrian traffic in the construction area and to prevent any pedestrian approaching into the areas of construction hazard, or any other dangerous area.

b. The Contractor shall be attentive to the needs of pedestrians that are visually or physically impaired, and the Contractor must be prepared at all times to assist in the safe and comfortable passage of these pedestrians.

c. The Contractor shall note that a number of existing utilities and services are located below the area of reconstruction and others in the near vicinity. The Contractor shall examine the site to identify potential problems associated with the accessibility, transportability and constructability of their proposed methods.

d. The Contractor shall, from time to time, adopt such approved construction or operating methods in carrying out the work as may be called for due to changing conditions that may be encountered during the progress thereof.

DS-10 Construction Survey and Layout

a. If the Drawings and Specifications requires the Contractor to perform the survey and layout of the project, the service shall be performed in accordance with "GS-3. Layout" and the Special Specifications. Sub-clauses pertaining to layout by the Owner are not applicable.

b. The survey and layout shall be performed or supervised by a competent surveyor with a minimum of five years related field experience (the "Surveyor"). The Contractor shall ensure the Surveyor attends a pre-construction meeting and other meetings as requested by the Contract Administrator.

c. The Contractor shall be responsible for ensuring the Surveyor prepares grade sheets and keeps proper digital records, notes and sketches of the survey and layout. A copy of the records (including but not limited to survey notes and sketches) shall be kept on site and accessible to the Contract Administrator at any time. The grade sheets shall be submitted to the Contract Administrator within 7 calendar days of production.

d. The Contractor shall produce a set of redlined Contract Drawings ("As-Built Field Record") marked with as-built information of the Project. The As-Built Field Record and all other records produced must be submitted to the Contract Administrator within 60 calendar days of Substantial Performance.

e. The compensation for the survey and layout shall be based on the lump sum amount for the tender item for the survey and layout identified in the Pricing Form. The lump sum amount for survey and layout shall not exceed 5 per cent of the Total Bid Price in the Pricing Form.

f. Eighty percent (80%) of the lump sum amount for the survey and layout shall be paid over the duration of the Contract in proportion to the value of the works completed as a percentage of the Total Bid Price. No payment will be made until the grade sheets are received by the Contract Administrator.

g. Twenty percent (20%) of the lump sum amount for the survey and layout shall be withheld and paid upon receipt and acceptance of the As-built Field Record. Should the Contractor fail to submit the As-built Field Record or if the submitted As-Built Field Record is not satisfactory to the Contract Administrator, this withheld amount may be used by the Contract Administrator to pay for an independent contractor to produce the As-built Field Record. If the cost to produce the As-built Field Record by the independent contractor exceeds the withheld amount, the Owner may exercise its right of set off under GC 5.10 – OWNER'S SET-OFF and deduct the additional cost from funds due and payable to the Contractor.

h. The Contractor must report to the Contract Administrator immediately any conflict, inconsistencies, errors, omissions, and/or discrepancies found between the Contract Drawing(s) and the existing physical conditions. Immediately upon becoming aware of such conflict, inconsistencies, errors, omissions and/or discrepancies, the Contractor shall stop survey and layout work until further directed by the Contract Administrator. The City shall not be responsible for any additional cost or time delay due to a failure of the Contractor or the Contractor's Surveyor to report in a timely manner such conflict or inconsistencies found, or due to a failure to suspend the survey and layout work pending direction from the Contract Administrator.

i. The Owner may conduct quality assurance verifications of the survey and layout as it deems necessary. The Owner's quality assurance process shall not relieve the Contractor of its responsibilities and obligations under this Contract. Any deficiency, omission or error identified by the Owner in the quality assurance process will be reported to the Contractor within two (2) Working Days. The Contractor shall verify the information provided by the Owner, and make adjustments or corrections where necessary. There shall be no additional compensation or extension of Contract Time for correction of the Contractor's deficiencies, omissions or errors.

j. Adjustments or corrections to the survey and layout required due to conflicts, inconsistencies, errors, omissions, and/or discrepancies between the Contract Drawing(s) provided by the Owner and the existing physical conditions will be compensated on a Time and Material basis in accordance with Schedule E - CHANGES IN THE WORK ON A TIME AND MATERIAL BASIS – LINEAR APPROACH. This compensation shall be limited to the cost of survey crew only.

DS-11 Disposal of Surplus Excavated Material and Removals

a. All surplus excavated materials, removals, grindings and all other debris, including that from sewer flushing and catch basin cleaning, shall be disposed of, off site. No separate payment shall be made for the costs associated with this work.

b. The City of Toronto will not make arrangements for the disposal of surplus materials or supply bills of lading.

c. The Contractor shall assume full ownership of the surplus excavated material and shall be solely responsible for its removal and disposal.

d. Stockpiling of excavated material within the City street allowance is not permitted. The Contractor shall dispose of all excavated material off site immediately upon removal. No additional payment will be made for costs incurred as a result of this requirement.

DS-12 Smog Alert Response Plans

a. The Contractor, when notified by the Consultant that the City's Smog Alert Response Plan has been implemented, shall, where applicable:

   i. suspend use of oil based products except for roadway line painting required to address safety concerns or to reduce traffic congestion;

   ii. suspend all pesticide spraying;

   iii. suspend grass cutting operations;

   iv. not allow refuelling during daytime hours;

   v. reduce equipment and vehicle idling as much as practical;

   vi. curtail the use of two-stroke engines as much as practical;

   vii. suspend normal street sweeping of all roadways during daytime hours except where there is an urgent need for clean-up, i.e. following a special event such as Caribana;

   viii. suspend the operation of loop cutting tar pots; and

   ix. suspend any non-essential planned traffic control device installation or modification work which will require lane closures or require complete deactivation of the traffic control device. Work that is required to address safety concerns or to reduce traffic congestion may continue.

b. Asphalt paving operations using SS-1 tack coat (water based) may continue.

c. A Smog Alert may be preceded by a Smog Watch. A Smog Watch is issued when there is a 50 percent chance that a smog day is coming within the next three (3) days. The Contractor shall not be entitled to any additional payment or extension of Contract Time due to the implementation of the Smog Alert Response Plans.

d. Notwithstanding the above, if it is necessary and the Contract Administrator ordered the suspension of paving operations, payment and/or extension of the Contract for the suspension of asphalt paving operations shall only be made if notification by the Chief Engineer and Executive Director or General Manager to suspend work is made in less than four hours prior to starting of such operations, and if such suspension has detrimentally impacted on the Contractor's work schedule. The Contractor shall provide supporting documentation identifying the impact and associated fair and reasonable costs in accordance with GC 2.6 – CONTRACTOR RECORDS and any delay in accordance with GC 7.5 - DELAYS.

e. Payment for this work, at actual costs incurred, shall be made under the appropriate provisional item(s) identified in the Pricing Form (if applicable), or otherwise and in accordance with Schedule E - CHANGES IN THE WORK ON A TIME AND MATERIAL BASIS – LINEAR APPROACH with the exception of any mark ups.

DS-13 Security and Construction Sign(s)

a. The Contractor shall be responsible for the security of the work of this Contract from the time the job site is turned over to him until all work has been completed.

b. The Contractor shall take all necessary precautions to ensure that the construction site does not pose a hazard to the public for the duration of the project. Appropriate safety and warning signs must be posted. All such site security measures shall be removed from the site at the completion of the project.

DS-14 Material and Truck Weighing

a. The City reserves the right to randomly verify the quantity of materials supplied in connection with this Contract. Prior to unloading of materials that are priced on a unit weight basis (“unit weight materials”), the weight tickets must be provided to the Contract Administrator (or in their absence, the City’s inspector). Material weight tickets that are not provided to the Contract Administrator or the City’s inspector prior to unloading will not be accepted later for payment.

b. When directed by the Contract Administrator or the City’s inspector, trucks carrying unit weight materials shall proceed immediately to a City’s weighing facility as specified by the Contractor Administrator or the inspector. After passing through the City’s weight scale and unloading the materials, the empty truck shall return to the same facility to verify the vehicle tare if so directed by the Contract Administrator or the City’s inspector.

c. Should the weight verification show that the verified weight of the material is less than what is shown on the Contractor’s weight ticket by more than 1.0 per cent, the payment for the affected load shall be made based on the weight measured by the City’s weighing facility.

d. City staff will also adjust the method of measurement for all following loads that are not weight-verified but have been delivered to the site before a new weight verification process can prove the Contractor had rectified the weight inconsistency. The weight of the following loads will be adjusted based on an adjustment factor determined from the most recently weight-verified load

e. The City will not compensate the Contractor for any cost associated with the weight verification process.

DS-15 Noise Regulations

a. The Contractor shall ensure the following:

b. Equipment shall be maintained in an operating condition that prevents unnecessary noise, including but not limited to proper muffler systems, properly secured components and the lubrication of all moving parts;

c. Idling of equipment shall be restricted to the minimum necessary for the proper performance of the specified work.

d. Where necessary, place noise attenuation devices (barriers) around Contractor's construction equipment.

DS-16 Traffic Signal and Street Lighting Installations

a. The Ontario Electrical Safety Code requires all "electrical installations", as defined in *Ontario Amendments to the Canadian Electrical Code Part I C22.1-02*, be inspected by the Electrical Safety Authority (ESA).

b. The Contractor shall file an application with the inspection department of the ESA 48 hours prior to the commencement of the work that requires the inspections. Information on inspection requirements and application for inspection can be found at www.esasafe.com/Contractors/ or by calling ESA at 1-877-esa-safe.

c. The Contractor shall provide an ESA issued "Certificate of Inspection" to the Contract Administrator prior to the Substantial Performance of the Contract.

d. The inspection fee shall be included in the appropriate bid items.

The provision of this list of designated substances in no way limits the Contractor's obligations as employer and, where applicable, as constructor under the OHSA, in particular those obligations with respect to designated substances. The Contractor, at no additional cost to the City, shall be responsible for making its own assessment as to which materials are designated substances at the project site, including any new material brought on site from time to time as construction progresses. Without limiting the foregoing, the Contractor shall not make any claims for delays or extra costs as a result of having to perform its obligations under the OHSA with respect to designated or hazardous substances.

Arsenic

Wood preservatives

Isocyanates

Sealants
Adhesives

Silica

Brick / Refractory Brick
Granite, Sandstone, Quartzite, Slate
Concrete. Concrete Block, Cement, Mortar
Mineral Deposits
Rock and Stone
Sand, Fill Dirt, Top Soil
Asphalt containing rock or stone

Asbestos

Asphalt
Transite Pipe
Old Mortar

Lead

Old Paint
Old Mortar
Old Water Pipes
Concrete
Solder
Sheet Metal

In addition to the designated substances above, the following hazardous materials may also be present:

Polychlorinated Biphenyls (PCBs)

Toronto Hydro Chambers
Toronto Hydro Conduits for Power Distribution

1 General

1.1 SUMMARY

.1 Section includes:

.1 Fire rated glazing and framing systems for installation as sidelights, windows, and wall sections in interior openings.

1.2 RELATED REQUIREMENTS

.1 Section 05 50 00 Miscellaneous Metals

.2 Section 07 84 00 Firestopping and Smokeseals

.3 Section 08 11 13 Steel Doors and Frames

.4 Section 08 70 00 Hardware

.5 Section 08 71 13 Automatic Door Operators

1.3 REFERENCE STANDARDS

.1 American Society for Testing and Materials (ASTM):

.1 ASTM A 1008/A 1008M - Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength, Low Alloy, and High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable; 2007.

.2 ASTM A 1011/A 1011M - Standard Specification for Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy and High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength; 2006b.

.3 ASTM E119: Methods for Fire Tests of Building Construction and Materials.

.2 American Welding Society (AWS)

.1 AWS D1.3 - Structural Welding Code - Sheet Steel; 2007

.3 Builders Hardware Manufacturers Association, Inc.

.1 BHMA A156 - American National Standards for door hardware; 2006 (ANSI/BHMA A156).

.4 Canadian Standards

.1 CAN-S101 Fire Endurance Tests of Building Construction and Materials

.2 CAN/ULC S104-M, “Fire Tests of Door Assemblies”

.3 CAN/ULC S106-M, “Standard Method for Fire Tests of Window and Glass Block Assemblies”

.5 National Fire Protection Association (NFPA):

.1 NFPA 80: Fire Doors and Windows.

.2 NFPA 251: Fire Tests of Building Construction & Materials

.3 NFPA 252: Fire Tests of Door Assemblies

.4 NFPA 257: Fire Test of Window Assemblies

.6 Underwriters Laboratories, Inc. (UL):

.1 UL 9: Fire Tests of Window Assemblies.

.2 UL 10 B: Fire Tests of Door Assemblies

.3 UL 10 C: Positive Pressure Fire Tests of Window & Door Assemblies

.4 UL 263: Fire tests of Building Construction and Materials

.5 UL-752 Ratings of Bullet-Resistant Materials

.7 American National Standards Institute (ANSI):

.1 ANSI Z97.1: Standard for Safety Glazing Materials Used in Buildings

.8 Consumer Product Safety Commission (CPSC):

.1 CPSC 16 CFR 1201: Safety Standard for Architectural Glazing Materials

.9 American Society of Civil Engineers (ASCE)

.1 ASCE 7 – Minimum Design Loads for Buildings and Other Structures; 2005

1.4 DEFINITIONS

.1 Manufacturer: A firm that produces primary glass, fabricated glass or framing as defined in referenced glazing publications.

1.5 SUBMITTALS

.1 Submit in accordance with Section 01 33 00.

.2 Product Data:

.1 Technical Information: Submit latest edition of manufacturer's product data providing product descriptions, technical data, Underwriters Laboratories, Inc. listings and installation instructions.

.3 Shop Drawings:

.1 Include plans, elevations and details of product showing component dimensions; framing opening requirements, dimensions, tolerances, and attachment to structure

.2 Provide templates for the location of embeds and anchor locations required for any adjoining work (if applicable).

.4 Structural Calculations (optional):

.1 Provide structural calculations sealed by a licensed professional engineer in the Province of Ontario; prepared in compliance with referenced documents and these specifications.

.5 Samples:

.1 Glass sample-as provided by manufacturer

.2 Sample of frame

.3 Verification of sample of selected finish

.6 Glazing Schedule: Use same designations indicated on drawings for glazed openings in preparing a schedule listing glass types and thicknesses for each size opening and location.

.7 Warranties: Submit manufacturer's warranty.

.8 Certificates of compliance from glass and glazing materials manufacturers attesting that glass and glazing materials furnished for project comply with requirements.

.1 Separate certification will not be required for glazing materials bearing manufacturer's permanent label designating type and thickness of glass, provided labels represent a quality control program involving a recognized certification agency or independent testing laboratory acceptable to authority having jurisdiction

1.6 QUALITY ASSURANCE

.1 Testing Agency Qualifications: Qualifications according to:

.1 International Accreditation Service for a Type A Third-Party Inspection Body (Field Services ICC-ES Third-Party Inspections Standard Operating Procedures, 00-BL-S0400 and S0401)

.2 International Accreditation Service for Testing Body-Building Materials and Systems

.1 Fire Testing

.1 ASTM Standards E 119

.2 CPSC Standards 16 CFR 1201

.3 NFPA Standards 251, 252, 257

.4 UL Standards 9, 10B, 10C, 1784, UL Subject 63

.5 BS 476; Part 22: 1987

.6 CAN Standards S 101, S 104, S 106

.2 Fire-Rated Wall Assemblies: Assemblies complying with ASTM E119 that are classified and labeled by UL, for fire ratings indicated, based on testing in accordance with UL 263, ASTM E119.

.3 Listings and Labels - Fire Rated Assemblies: Under current follow-up service by Underwriters Laboratories maintaining a current listing or certification. Label assemblies accordance with limits of manufacturer's listing.

1.7 DELIVERY, STORAGE AND HANDLING

.1 Deliver, store and handle under provisions specified by manufacturer.

1.8 PROJECT CONDITIONS

.1 Obtain field measurements prior to fabrication of frame units. If field measurements will not be available in a timely manner coordinate planned measurements with the work of other sections.

.1 Note whether field or planned dimensions were used in the creation of the shop drawings.

.2 Coordinate the work of this section with others effected including but not limited to: other interior components and door hardware beyond that provided by this section

1.9 WARRANTY

.1 Provide the standard five (5) year manufacturer warranty.

2 Products

2.1 MANUFACTURERS - (ACCEPTABLE MANUFACTURERS/PRODUCTS]

.1 Manufacturer Glazing Material: "Pilkington Pyrostop®" fire-rated glazing as manufactured by the Pilkington Group and distributed by Technical Glass Products, 8107 Bracken Place SE, Snoqualmie, WA 98065 (800-426-0279) fax (800-451-9857) e-mail sales@fireglass.com, web site http://www.fireglass.com

.2 Frame System: "Fireframes® Aluminum Series" fire-rated frame system as manufactured and supplied by Technical Glass Products, 8107 Bracken Place SE, Snoqualmie, WA 98065 (800-426-0279) fax (800-451-9857) e-mail sales@fireglass.com web site http://www.fireglass.com

2.2 PERFORMANCE REQUIREMENTS

.1 System Description:

.1 Steel fire-rated glazed wall and/or window system, dual aluminum cover cap format

.1 Face widths available:

.1 50mm (2")

.2 Custom extruded aluminum cover caps

.2 Duration – Windows Capable of providing a fire rating for 45 minutes.

.3 Duration – Walls: Capable of providing a fire rating for 60 minutes.

.2 Delegated design: For the performance requirements listed below requiring structural design provide data, calculations and drawings signed and sealed by an engineer licensed in the Province of Ontario.

2.3 MATERIALS - GLASS

.1 Fire Rated Glazing: Composed of multiple sheets of Pilkington Optiwhite™ high visible light transmission glass laminated with an intumescent interlayer.

.2 Impact Safety Resistance: ANSI Z97.1 and CPSC 16CFR1201(Cat. I and II).

.3 Properties Interior Glazing

Property	
Fire Rating	45 minute
Manufacturer's designation	45-200
Glazing type	single
Nominal Thickness	3/4" (19mm)
Weight in lbs/sf	9.2
Daylight Transmission	86
Sound Transmission Coefficient	40dB

.4 Logo: Each piece of fire-rated glazing shall be labeled with a permanent logo including name of product, manufacture, testing laboratory (UL), fire rating period, safety glazing standards, and date of manufacture.

.5 Glazing Accessories: Manufacturer's standard compression gaskets, standoff, spacers, setting blocks and other accessories necessary for a complete installation.

2.4 MATERIALS –ALUMINUM FRAMES

.1 Aluminum Framing System 60 min.

.1 Steel Frame — The steel framing members are made of two halves, nom. 1.9 in. wide (48.3 mm) with a nom. minimum depth of 1.38 in. (35 mm) with lengths cut according to glazing size.

.2 Aluminum Trim — Supplied with the steel framing members. Nom. 2 in. (50.8 mm) wide with a nom. depth of 1.54 in. (39 mm) with lengths cut according to glazing size.

.3 Framing Member Fasteners — Supplied with the steel framing members. Screws are M6 x16mm Button Head Socket Cap Screws for frame assembly and #6 x 1" Pan Head Sheet Metal Screws for door installation.

.4 Glazing Gasket — Supplied with the steel framing members. Nom. 3/4 in. (19 mm) by 3/16 in. (4.5 mm) black applied to the steel framing members to cushion and seal the glazing material when installed.

2.5 DOOR HARDWARE

.1 Manufacturer's heavy duty hardware units in sizes and types as required to meet fire rated entrance use as indicated on Drawings.

.1 Provide door hardware in accordance with the requirements of this Section; using products that are recommended and supplied by entrance system manufacturer; in accordance with referenced standards, meeting requirements for description, quality, type, and function listed in hardware schedule.

2.6 FABRICATION

.1 Obtain reviewed shop drawings prior to fabrication.

.2 Fabrication Dimensions: Fabricate fire-rated assembly to field dimensions.

.3 Factory prepared, fire-rated steel door assemblies by TGP to be prehung, prefinished with hardware preinstalled for field mounting.

.4 Field glaze door and frame assemblies.

2.7 FINISHES, GENERAL

.1 Comply with NAAMM's "Metal Finishes Manual for Architectural and Metal Products" for recommendations for applying and designating finishes.

.2 Finish frames after assembly.

.3 Appearance of Finished Work: Variations in appearance of abutting or adjacent pieces are acceptable. Noticeable variations in the same piece are not acceptable.

2.8 FINISHES

.1 Finish after fabrication.

.2 Appearance of Finished Work: Variations in appearance of abutting or adjacent pieces are acceptable. Noticeable variations in the same piece are not acceptable.

.3 Anodized Finishes

.1 Finish designations prefixed by AA comply with the system established by the Aluminum Association for designating aluminum finishes.

.2 Class II, Clear Anodic Finish: AA-M12C22A31 (Mechanical Finish: nonspecular as fabricated; Chemical Finish: etched, medium matte; Anodic Coating: Architectural Class II, clear coating 0.010 mm or thicker) complying with AAMA 611.

.4 Steel (Concealed):

.1 Hot-dip galvanized, or zinc rich paint.

.5 Isolate where necessary to prevent electrolysis due to dissimilar metal-to-metal contact or metal-to-masonry and concrete contact. Use bituminous paint, butyl tape or other approved divorcing material.

.1 Bituminous Paint: Cold-applied, asphalt-mastic paint complying with SSPC-Paint 12 requirements except containing no asbestos; formulated for 0.762-mm thickness per coat.

3 Execution

3.1 EXAMINATION

.1 Site Verification of Conditions: Examine substrates and members to which the work of this section attaches or adjoins prior to frame installation are acceptable for product installation in accordance with manufacturer’s instructions. Provide openings plumb, square and within allowable tolerances. The manufacturer recommends 3/8 inch shim space at all walls

.2 Notify Architect of any conditions which jeopardize the integrity of the proposed fire wall / door system.

.3 Do not proceed until such conditions are corrected.

3.2 INSTALLATION

.1 Install as per manufacturer’s instructions.

.2 General:

.1 Erect fire rated aluminum framed entrances plumb, level and square, in correct relation to work of other sections, within a maximum non-cumulative deviation of 1/8" per 12'-0" length of member, and with members accurately fitted and aligned at joints and intersections.

.2 Anchor system to building structure, adjusting as required to meet erection tolerances and secure to prevent movement other than that which is expected due to structural deflection and creep and thermal expansion and contraction.

.3 Provide all devices and components required for erection of system.

.4 Use concealed fastenings only.

.5 Touch up steel anchoring components, after installation, with zinc rich paint.

.6 Provide aluminum flashings, fillers, covers and sealants indicated and as required to render system weather tight and to meet specified performance criteria. Ensure effective seal at laps, end joints and changes of direction.

.7 Seal joints between storefront framing system and adjacent building elements, and between frames, sills and other materials. Caulk inside and outside, with sealant as specified herein.

.8 Install all door hardware on doors. Test all doors on completion of installation and adjust as required for smooth and efficient operation.

.9 Completed installation shall be of adequate strength to support operating entrance doors, and wind loading as specified without glass shaking or vibrating when entrance doors are in use.

.3 Glazing:

.1 Install glass types as indicated in Section 08 80 00.

.2 Size glass units to accurately fit openings with appropriate clearance all around.

.3 Identify glazed openings, mark each light of glass. Indicate presence of glass.

.4 Replace all damaged or broken glass at no expense to Owner, prior to completion of work. Remove all broken glass from premises.

.5 Locate and install setting blocks and spacers according to glass manufacturer's directions. Centre and space each piece of glass on premoulded neoprene rubber spacers. Provide minimum of two spacers on each edge of each piece of glass and four where dimension exceeds 4’. Use spacers of size to accurately fit each thickness of glass.

.6 Clean glass and metal surfaces to present clean, dry, grease and oil free surfaces to receive glazing tapes, gaskets or seals.

.7 Glazing to be undertaken at temperatures recommended by manufacturer of glazing materials.

.8 Provide sealed double glazed units at all locations.

3.3 REPAIR AND TOUCH UP

.1 Anodized Finishes:

.1 Protect the anodized finish from harsh chemicals such as concrete/mortar or muriatic acid/brick wash. If reasonable care is taken during handling and high and low pH chemicals can be avoided, repair and/or touch-up of an anodize finish will not be needed.

.2 Some rub marks on an anodized surface can be removed with a mild abrasive pad such as a Scotch-Brite pad prior to touch up painting.

.3 Touch-up paint should be used even more sparingly over anodize. Only the visible raw aluminum in the scratch or gouge should be touched up with a matching paint.

.2 Remove and replace glass that is broken, chipped, cracked, abraded, or damaged.

3.4 PROTECTION AND CLEANING

.1 Protect glass from damage immediately after installation by attaching crossed streamers to framing held away from glass. Do not apply markers to glass surface. Remove nonpermanent labels, and clean surfaces.

  .1 Do not clean with astringent cleaners. Use a clean “grit free” cloth and a small amount of mild soap and water or mild detergent.

.2 Protect glass from contact with contaminating substances resulting from construction operations, including weld splatter. If, despite such protection, contaminating substances do come into contact with glass, remove them immediately as recommended by glass manufacturer.

END OF SECTION

PART 1 - GENERAL

1.1 General Requirement

.1 Comply with the applicable requirements of Division 1.

1.2 Work Furnished and Installed

.1 Plain and reinforced concrete
.2 Formwork
.3 Grout under base plates

1.3 Work Installed but Furnished by Other Sections

.1 Anchor bolts	Section 05 12 23
.2 Embedded items, sleeves, boxes	Divisions 22, 23 and 26
.3 Shelf angles, wall plates and base plates anchored into concrete members	Section 05 12 23

1.4 Work Furnished but not Installed

.1 Concrete and reinforcement bars to masonry trade.

1.5 Related Work Specified Elsewhere

Concrete Formwork	Section 03 10 00
Concrete Reinforcement	Section 03 20 00
Structural Steel	Section 05 12 23
Steel Deck	Section 05 31 00
Insulation	Section 07 20 00
Excavating, Trenching and Backfilling	Section 31 23 10

1.6 Codes and Standards

.1 Do cast-in-place concrete work in accordance with the OBC 2012 designated Editions of CAN/CSA-A23.1 and testing in accordance with CAN/CSA-A23.2, unless otherwise noted.

.2 Comply with the requirements of the Ontario Building Code 2012, and The Occupational Health and Safety Act, and Regulations for Construction Projects, latest issue including all amendments and revisions.

.3 Keep the following references in the project field office: CAN/CSA-A23 and CAN/CSA-A23.2, Reinforcing Steel Institute of Ontario - Manual of Standard Practice and CAN/CSA-S269.3.

1.7 Source Quality Control

.1 Before concrete is placed, supply the Consultant with details of the mix proposed for each class of concrete; giving proportions of cement, coarse In addition, fine aggregate, type and amount of admixtures or air entraining agents and water-cement ratio. Certification that compressive strength, slump, air-entrained content and other specified properties will be met.

.2 Provide Consultant with mill test reports properly correlated to the reinforcement.

1.8 Quality Assurance

.1 Obtain concrete from a member of the Ready-Mixed Concrete Association of Ontario that has been issued a seal of Special Quality Concrete who can guarantee to produce concrete with the coefficient of variation less than 12 percent.

.2 Engage an organization with at least 5 years of specialized experience to undertake floor finishing. Submit substantiating references if asked.

1.9 Site Conditions

.1 Determine any potential interference with existing services and protect from disruption and damage.

PART 2 - PRODUCTS

2.1 Concrete Material

.1 Cement: Normal Portland cement to CAN/CSA-A5, Type 10.

.2 Use only one brand of cement for architectural concrete.

.3 Mixing water: Clear and potable to CAN/CSA-A23.1

.4 Fine aggregate: Natural sand to CAN/CSA-A23.1

.5 Coarse aggregate: crushed stone or gravel to CAN/CSA-A23.1, suitable for NBC type N concrete. Maximum size 20 mm except 12 mm for toppings.

.6 Low density aggregate for light weight concrete to CAN/CSA-A23.1.

2.2 Admixtures

.1 Comply with manufacturer's instructions.

.2 Obtain admixtures from single source.

.3 Air Entraining admixture: to CAN/CSA-A266.1

.4 Non-Retarding Water Reducing Agent: CAN/CSA-A266.2, Type WN

.5 Corrosion inhibitor: Provide DCI or MCI concrete admixtures as per CAN/CSA-S413 recommendations

2.3 Grouts

.1 Premixed grout: Minimum strength 40 MPa at 28 days. Install in accordance with manufacturer's recommendations.

.2 Non premixed dry packed grout: 1:3 (cement: sand) and minimum water. Minimum strength 30 MPa at 28 days.

.3 Use only non-metallic grout in exposed surfaces subject to staining.

2.4 Accessories

.1 Comply with manufactures' instructions.

.2 Water-stop:Polyvinyl Chloride, PVC To CSGB 41-GP-35M, Types 1 and 3

.3 Adjustable wedge anchor - Peerless Wedge, anchor insert malleable Acrow-Richmond, iron Lintel Anchor, Superior Concrete

.4 Drilled concrete anchor - Acrow-Richmond Wej-it or Parabolt Hilti Kwik-bolt

.5 Non-slip nosing insert - Fine Aluminium oxide strips, 6 mm wide x 10 mm deep.

.6 Premoulded joint filler - Bituminous impregnated fibre board to ASTM D1751

.7 Membrane adhesive- As recommended by membrane manufacturer.

.8 Saw cut joint filler - Sika Canada, Loadflex

2.5 Curing-Sealing Compound

.1 Clear lacquer type liquid conforming to ASTM C309. It shall not darken or discolour concrete surface, not impair bonding of any material laid over surface and shall be compatible with such materials.

2.6 Floor Surface Hardener

.1 Comply with the manufacturer's instructions.

.2 Non-metallic, natural grey colour:

Mastercon (Curing-sealing compound-Masterseal)	- Master Builders Co. Ltd.
Diamag 7 (Curing-sealing compound-Clear Florseal)	- Sika Canada

2.7 Proportioning

.1 Conform to CAN/CSA-A23.1, Section 14. Select mix proportions to provide the specified strength, workability and durability.

.2 Minimum cement content for exposed concrete slabs:

Foot traffic only	- 285 kg/m3
Vehicular traffic with rubber wheels	- 320 kg/m3

.4 Maximum water: cement ratio for concrete subject to freezing and thawing:

- CSA Class C or Class A, according to exposure

.5 Use the following chemical admixtures:

All concrete	- Water Reducing Agent
Concrete subject to cycles of freezing and thawing	- Air Entraining Agent

Obtain approval of the Architect for the use of other admixtures prior to use.

Main parking garage ramp. Corrosion inhibiter

.6 Maximum Slumps:

Foundations, wall and slabs	- 70 mm
Toppings	- 60 mm
Other concrete	- 80 mm
Parking structures	- 40 mm

2.8 Production

.1 Use ready-mixed concrete.
.2 Use corrosion inhibitors for ramped slab concrete as per CSA S413-07 recommendations.
.3 Heat concrete and deliver at a temperature between +13°C and +27°C whenever outdoor temperature is less than +5°C.

PART 3 - EXECUTION

3.1 Workmanship

.1 Comply with the requirements of CAN/CSA-A23.1 and the specific requirements of the Contract.

.2 Ensure that no water is present on foundation beds where footings and other concrete work is to be placed. Place concrete only on frost-free ground. Remove previously frozen bearing surfaces.

.3 Obtain Consultant's approval prior to placing concrete.

.4 Ensure that reinforcement and inserts are not disturbed during concrete placement.

.5 Do not place load on new concrete until authorized by Consultant.

.6 Bring to the attention of the Consultant any defects or deficiencies in the Work which may occur during construction together with a proposal for remedy. Consultant will decide what corrective action may be taken.

3.2 Records

.1 Maintain accurate records of poured concrete items to indicate date, location of pour, quality, air temperature and test samples taken. Record the date of removal of each section of formwork.

.2 Record on drawings the founding elevations of all footings and variations of footing work from that indicated on drawings.

3.3 Cooperation

.1 Cooperate with all engaged on the Project. Exchange with related trades, shop drawings and other data required to coordinate and schedule the Work. Notify other trades as to when items which are to be installed by they are to be set and protect these items after installation.

.2 Give the Consultant at least 24 hours advance notice of the time when completed reinforcement will be ready for review.

.3 Supply and install grout for base and bearing plates. Coordinate installation with the Structural Steel trade. Grout shall completely fill space between plates and supports.

.4 Cooperate with any trade applying finishes to concrete surfaces to obtain a surface which will have adequate bond. Provide chases where required.

3.4 Examination of Work

.1 Do not begin operations before making a thorough examination of existing conditions and the Work of related trades. Report inconsistencies before proceeding.

3.5 Inspection and Testing

.1 The Consultant will appoint an independent inspection and testing agency to undertake concrete strength tests.

.2 Pay for the cost of inspection from the Cash Allowance, as directed by Consultant.

.3 Assist the agency in its work. Notify it as to the concreting schedule. Provide concrete samples and standard test cylinders.

.4 Laboratory curing and testing of samples will be carried out in accordance with the applicable CSA Standards. The agency will report to the Consultant with copies to Structural Subconsultant, the Contractor and the Municipal Authorities. Reports will be made on a form similar to Appendix C of CAN/CSA-A23.2 stating the location of concrete to which tests relate and commenting on abnormal results and conditions.

.5 Provide a group of three cylinders for each standard strength test. One specimen will be tested at 7 days and two at 28 days.

.6 Provide one additional site cured cylinder for testing at 7 days when concrete is placed under cold weather conditions.

.7 Take samples at the discharge end of the pipe when concrete is pumped.

3.6 Rejected Work

.1 Do not deliver to the site materials which are known not to meet the requirements of the Specifications. If rejected after delivery they shall be immediately removed.

.2 Where review reveals materials or workmanship which appear to have failed to meet the specified quality or tolerances, the Consultant shall have the authority to order additional curing; to have tests made of in-situ concrete, concrete cores, reinforcement or other materials; to order a structural analysis of the existing elements and to load test the structure. All such work will be carried out in order to assist in determining whether the structure may, in the opinion of the Architect be accepted, with or without strengthening or modification. Testing shall meet the requirements of the Ontario Building Code. All expense incurred shall be chargeable to the Contractor regardless of the results.

3.7 Quality Control On-Site

.1 Make all required field measurements.

.2 Do not close deep forms until reinforcement has been reviewed.

.3 Ensure that reinforcement is kept free from dirt, grease, loose mill scale and rust. Ensure that reinforcement is complete, adequately tied and properly positioned for cover in advance of the time scheduled for casting concrete.

.4 Make available to the Consultant, documents to verify that the concrete supplier is qualified to supply ready-mixed concrete and on to review proposed concrete mix design.

.5 Make one standard strength test for each 100 m^3, but not less than one test, for each class of concrete placed each day. Store cylinders in metal lined curing boxes maintained at a temperature of +10`C until shipped to the testing laboratory. Store additional cylinder required for cold weather or hot weather conditions adjacent to Work for 7 days.

.6 Conduct one standard air entrainment test for each 50 m^3 of air-entrained concrete or portion thereof placed each day.

.7 Make slump tests with each standard strength test and when so directed by Consultant.

3.8 Joints

.1 Construction Joints:

.1 Provide construction joints where specified or shown on the drawings. Locate and make other joints so as not to impair the required strength of the structure. Joints are subject to the review of the Architect.

.2 Unless otherwise shown, maximum distances between construction joints are:

Walls - 10 m, or 20 m alternating with control joints at same spacing.
Slabs on grade - 7 m, or 20 m with sawcut joints at 5 m centres.

.2 Isolation Joints:

.1 Provide 6 mm thick premoulded asphalt impregnated joint filler of the same depth as the thickness of the concrete wherever slabs on grade abut foundation walls, columns and piers.

.3 Saw-cut Joints:

.1 Saw-cut joints in slabs on grade should generally occur on column lines and on intermediate lines which result in panels of approximately square sections. Sections shall not exceed 5 m in length or width.

.2 Make saw-cuts 3 mm wide and 30 mm deep as soon as the concrete can be cleanly cut and after shrinkage cracks can form. Not less than 21 days after casting, fill all saw cuts with polysulfide joint filler. Joints shall be clean and dry when filled.

3.9 Placement of Concrete

.1 Remove water from excavations before placing concrete therein.

.2 Convey concrete from mixer to place of final deposit by methods which will prevent aggregation of materials and change of concrete qualities. Time for this operation shall not exceed 30 minutes. Deposit concrete as close as possible to its final position.

3.10 Slabs on Grade

.1 Wet the subgrade surface by sprinkling before placement of concrete.

.2 See Drawings for thickness of concrete and slab reinforcing.

.3 Refer to Architectural Drawings for slab depressions, slopes and finishes. Slope floors to drains.

3.11 Surface Finishing

.1 Honeycomb:

In locations where the repair of honeycomb is approved, cut out defective areas and fill the space with a cement mortar of the same materials as the concrete. Incorporate a liquid latex bonding agent into the mix. Apply in layers not exceeding 25 mm in thickness.

3.12 Curing and Protection

.1 Beginning immediately after placement, protect concrete from premature drying, sunshine, excessively hot or cold temperatures, and mechanical injury. Maintain at a relatively constant temperature for as long as is required for hydration of the cement and curing of the concrete. Provide adequate moisture under dry conditions by wetting subgrade and surrounding construction as appropriate.

.2 Minimize moisture loss from surfaces placed against wooden forms, or plastic and metal forms exposed to heating by the sun, by keeping the forms wet until they can be safely removed. If forms are removed in less than 7 days, curing shall be continued by one of the wet curing methods specified for surfaces not in contact with forms.

.3 Select curing methods best suited to the ambient conditions in which the structure is being constructed.

.4 Cure concrete surfaces not in contact with forms by one of the following methods:

  .1 Pounding or continuous sprinkling.

  .2 Application of absorptive covering kept continuously wet.

  .3 Application of fog spray followed by a covering of curing paper lapped 150 mm and held down at all edges.

.4 Application of a curing-sealing compound, where permitted, immediately after disappearance of surface water sheen. Do not use a compound unless it is compatible with any material which may be applied to or laid over the concrete surface.

.5 Curing methods based upon keeping surfaces wet shall continue for at least seven days. Prevent intermittent drying of surfaces.

.5 Do not pile, store or transport materials over slabs until concrete has been in place for at least 7 days.

END OF SECTION

PART 1 - GENERAL

1.1 GENERAL REQUIREMENT

.1 Comply with the applicable requirements of Division 1.

1.2 RELATED WORK

.1	Concrete formwork	Section 03 10 00
.2	Cast-in-place concrete	Section 03 30 00

1.3 WORK FURNISHED BUT NOT INSTALLED

.1 Reinforcement bars to masonry trade.

1.4 CODES AND STANDARDS

.1 Do reinforcing work in accordance with the Ontario Building Code 2012 designated editions of CAN/CSA-A23.1 and testing in accordance with CAN/CSA-A23.2, and the Manual of Standard Practice of Reinforcing Steel Institute of Canada.

1.5 TEST REPORTS

.1 Upon request, provide Consultant with certified copy of mill test report of steel supplied, showing physical and chemical analysis, at least 2 weeks prior to commencing reinforcing work.

PART 2 - PRODUCTS

2.1 MATERIALS

.1 Reinforcing steel: Billet steel, grade 400, deformed bars to CSA G30.12, unless indicated otherwise.

.2 Cold-drawn annealed steel wire ties: to CSA G30.3.

.3 Deformed steel wire fabric for concrete reinforcement: to CSA G30.14.

.4 Welded steel wire fabric: to CSA G30.5. Provide in flat sheets only.

.5 Welded deformed steel wire fabric: to CSA G30.15. Provide in flat sheets only.

.6 Chairs, bolsters, bar supports, spacers: to CAN/CSA-A23.1 and adequate for strength and conditions.

.7 Mechanical splices: subject to approval of Consultant.

2.2 FABRICATION

.1 Fabricate reinforcing to CAN/CSA-A23.1.

.2 Fabrication tolerances for reinforcing steel to RSIO Manual of Standard Practice.

.3 Obtain Consultant's approval for locations of reinforcement splices other than shown on steel placing drawings.

.4 Do not bend or weld epoxy coated bars after coating.

.5 Ship bundles of bar reinforcement clearly identified in accordance with bar list.

PART 3 - EXECUTION

3.1 HANDLING AND FIELD BENDING

.1 Do not field bend reinforcement except where indicated or authorized by Consultant.

.2 When field bending is authorized, bend without heat, applying a slow and steady pressure.

.3 Replace bars which develop cracks or splits.

3.2 WORKMANSHIP

.1 Comply with the requirements of CAN/CSA-A23.1 and the specific requirements of the Contract.

.2 Comply with the requirement of CSA-S413 for all work in the underground parking slabs and ground floor at parking areas and drive ways.

.3 Ensure that reinforcement and inserts are not disturbed during concrete placement.

3.3 PLACEMENT OF REINFORCING

.1 Store reinforcement on racks or skids so that it will be protected from dirt and maintained in its fabricated form.

.2 Use only approved shop drawings and the Structural Drawings for placing of reinforcement. Report discrepancies to the Architect before proceeding.

.3 Bend all bars cold. No field bending or tack welding is allowed.

.4 Before placing, remove all loose scale, dirt, oil or other coatings which would reduce bond. Place reinforcement within the specified tolerances and secure in position by the use of chairs, spacers and hangers. Tie securely together using annealed wire to prevent displacement during concrete placing and vibrating. Use non-corrosive tie wire for architectural concrete. Turn the ends of ties towards the interior of the concrete.

.5 Position reinforcing for exposed concrete using snap-on plastic positioners and chairs with plastic tipped legs of the same color as the concrete. Use concrete chairs for slabs on grade and footings.

.6 No splicing of reinforcement is permitted other than shown on the Structural Drawings. Do not cut reinforcement to permit placing of embedded items.

.7 Lap end cross wires of welded wire mesh at least 200 mm.

.8 Do not cut reinforcement, either before or after concrete is placed.

.9 Do not cut or displace bars that interfere with sleeves or other openings without the approval of the Consultant.

3.4 INSPECTION AND TESTING

.1 Consultant will appoint an independent inspection and testing agency to undertake inspection of reinforcement steel.

.2 This inspection shall be paid for from the Cash Allowance, as directed by the Consultant.

END OF SECTION

PART 1 - GENERAL

1.1 GENERAL REQUIREMENT

.1 Comply with the applicable requirements of Division 1.

.2 Furnish all labour, materials, equipment, tools and incidentals necessary for the complete design, installation, maintenance and removal of concrete formwork as required by the drawings and/or as herein specified, including: falsework, bracing, supports, forms, ties, templates, blockouts, sleeves, placing only of anchor bolts and all inserts related to the curtain wall details. Supply and placing of all inserts imbedded in the concrete, including water stops.

1.2 RELATED WORK

.1 Concrete Reinforcement	Section 03 20 00
.2 Cast-In-Place Concrete	Section 03 30 00

1.2 REFERENCE STANDARDS

.1 Do concrete formwork in accordance with CAN/CSA-S269.3.

.2 Comply with the requirements of the Ontario Building Code 2012, and The Occupational Health and Safety Act, and Regulations for Construction Projects, latest issue including all amendments and revisions.

.3 Refer to current edition of all Regulations and By-laws.

1.3 APPROVAL

.1 Obtain Consultant's approval of all forms for exposed concrete work prior to placing concrete. Obtain written approval of Consultant for all mechanical and electrical openings in exposed work.

1.4 SHOP DRAWINGS

.1 Prepare formwork shop drawings to clearly illustrate all materials and details of work, complete and to adequate scale. Comply with the requirements of Section 01 33 00.

.2 Drawings shall contain a description of the methods of shoring and reshoring floor slabs and other horizontal members, shoring for vertical walls, locations and methods of forming for expansion, construction and control joints, methods of stripping and all other relevant data.

.3 Drawings shall show all relevant dimensions. Indicate location of construction joints, inserts, openings, sleeves and the like and for all exposed work show location of clean-outs for columns and high walls.

.4 Drawings shall show formwork pattern and material, special details for architectural finishes, pattern and type of formwork ties for exposed work.

.5 All formwork drawings shall bear the stamp and signature of a qualified professional engineer.

.6 Submit copies of formwork shop drawings to the Consultant for approval. No work indicated on such drawings shall be done until these drawings have been approved by engineer.

.7 Formwork shop drawings will be checked for general arrangement only and not for figured dimensions and approval shall in no way relieve the contractor of his responsibility for carrying out the work in accordance with the contract documents and intent.

.8 After review, drawings will be returned to the Contractor stamped to show one of the followings:

Reviewed	-	Released for fabrication
Reviewed as Noted	-	Released for fabrication after revisions noted are made, submit final record print.
Revise and Resubmit	-	Correct and resubmit for review prior to fabrication.

.9 Keep on site at all times a set of shop drawings bearing the review stamps of the Consultant and the Structural Sub-consultant and use only these drawings and the Structural Drawings to place reinforcing steel.

Neatly mark on the Structural Drawing changes issued during the course of construction.

1.5 EXAMINATION OF DRAWINGS

.1 Check all dimensions and details given on the structural drawings against those on the architectural drawings. Have discrepancies, if any, corrected before starting construction of formwork.

PART 2 - PRODUCTS

2.1 FORMWORK MATERIAL

.1 Quality and strength of formwork material shall comply with the requirements of CAN/CSA-S269.3.

.2 Use formwork materials which are new at start of Work except that, for unexposed locations such as foundations, used materials may be substituted.

.3 Plywood, wood and formwork materials to CAN/CSA-O86.

.4 False material to CAN/CSA-S269.1.

.5 Tubular column forms: round, spirally wound laminated fibre forms, internally treated with release material.

.6 Grooves, reglets and chamfers: White Pine selected for straightness and accurately dressed to size.

.7 If a smooth architectural finish to concrete is required, specify form materials such as high density overlaid plywood to CSA O121 or other special materials to achieve the desired concrete finish.

2.2 ACCESSORIES

.1 Water-stops: Plastic type, 150 mm wide minimum, weighing 1.75 Kg/m for solid sections and 1.5 Kg/m for sections with one split leg, made of continuous extruded virgin polyvinyl, capable of 300% elongation.

.2 Non-slip nosing insert: Fine Aluminium oxide strips, 6 mm wide x 10 mm deep.

.3 Joint Tape: Non-staining, water impermeable, self-release.

2.3 FORM TIES

.1 Snap-ties and ties capable of acting as both tie and spread. Ties for exposed work shall be hot-dip galvanized equipped with snap off metal ties to provide for a break point 40 mm from the concrete surface.

.2 Ties shall have a minimum working strength of 1300 kg. Ties for exposed work shall be equipped with 40 mm high wood or other non-staining removable cones, or as selected by the Consultant. Use of Wire ties is not permitted.

2.4 FORM RELEASE AGENT

.1 Non-staining type, based on chemical reaction between the release agent and alkali content of concrete. Oil based lubricants may not be used without engineer's approval.

2.5 SAMPLES

.1 Submit samples of form ply, boards, plastic, coatings, and release agents, form ties and cones and other relevant material affecting the concrete finish

PART 3 - EXECUTION

3.1 TYPE, STRENGTH, RIGIDITY AND ALIGNMENT OF FORMS

.1 Forms shall be of sufficient strength and rigidity to support all concrete and construction loads and wind, taking into account proposed rate and method of pouring the concrete so that the resultant finished concrete shall conform to the shapes, lines and dimensions of the members shown on drawings.

.2 Forms shall produce straight dense surfaces, free from honeycombs, bulges and depressions.

.3 Special cambers if any shall be shown on drawings.

.4 Carefully lay out concrete forms from batter boards and fine wire check lines. Check and correct wedging and shoring both horizontally and vertically as required during the placing of concrete. When concrete is being placed in a wall provide at least one readily accessible check wire parallel to the direction of the wall.

Assign at least one competent carpenter to check continually the alignment during the operation of placing concrete. Take special care in the alignment of exterior exposed columns and beams.

.5 Form sides of footings unless otherwise noted on the Structural Drawings.

.6 Do not permit loads from formwork to be transmitted to adjacent existing structure.

.7 Apply a form coating and release agent uniformly to the contact surface of formwork panels before reuse. Seal all lumber in forms for architectural concrete prior to use.

3.2 SHORING AND BRACING

.1 Provide bracing to ensure the stability of the formwork at all times.

.2 Prop or strengthen all previously constructed parts liable to be overstressed by construction loads.

.3 Place shores in successive storeys of a structure directly over those below.

.4 Design forms to allow stripping without removal of the principal shores, where these are required to remain in place.

.5 Keep shoring or re-shoring in position until concrete has reached its 28 day strength.

3.3 JOINTS IN FORMS

.1 Make form joints tight in order to prevent leakage of grout, particularly at corners and at all joints between board and panels used for exposed work. Do not use tape or the like for exposed work.

.2 Clean all edges and contact surfaces before erection.

3.4 HORIZONTAL CONSTRUCTION JOINTS

.1 Make all horizontal construction joints on exposed surfaces straight and level.

3.5 WATERSTOPS

.1 Install water-stops in all expansion, construction and other joints where shown on the drawings.

.2 Set and secure water-stops in forms so that their position is maintained during placing of concrete.

3.6 CORNER BEVELS

.1 Construct corners as detailed on architectural drawings. Unless otherwise shown, exposed columns, walls and beams must have a 20mm chamfer and all other corners shall be square and sharp

3.7 CLEANING FORMS

.1 Clean out forms before pouring concrete. Provide all inaccessible portions of formwork with clean-out and inspection ports. Column forms shall have a 150 x 250 mm port built in at the bottom of the column so arranged that it can be removed and the inside space cleaned of all debris before casting concrete. The port shall have a closure that can be put in place and held under concrete pressure without distortion of members. Provide similar ports in the base of walls and other deep members at about 3 m intervals.

3.8 SLEEVES AND PENETRATIONS

.1 Install sleeves only as shown on drawings or as directed by the Consultant.

.2 Form all multiple services of through-penetrating items to meet manufacturers and other specified fire stopping requirements as per related Sections/Documents and as outlined below.

.3 Bear all extra costs incurred as a result and in the event of oversized joints (larger than specified).

.4 The following regulates the sizing of service penetrations to be fire stopped in an effort to standardize and minimize penetration sizes:

.1 Sleeve single, circular penetrates by Divisions 15 and 16 respectively.
Multiple penetrations of circular penetrates shall be considered such, if the circular penetrates are no further than 100 mm (4") apart. Such multiple penetrations shall be carried out by this Section where erecting the fire separations by forming an open, square or rectangular, box around the multiple penetrates. This box, or frame, shall have a maximum 25 mm (1") clearance around the outer penetrates.

.2 Create penetrations with square penetrates in the same manner as the above mentioned multiple, circular penetrates.

The only difference is that the maximum clearance between penetrant and penetration shall be a maximum of 50 mm (2").

.5 An exception to this is the fire dampers, which require a design specific clearance around them.

3.9 SURFACE TREATMENT OF FORMS

.1 Use untreated forms where concrete is to receive stucco or plaster finish. Soak inside surfaces of these forms subject to shrinkage or absorption of water and keep continuously wet prior to pouring of concrete.

.2 Treat plywood and board forms, where concrete is to be exposed to view, with a non-staining form release agent or with an approved type of lacquer or plastic before each use of forms. Spray apply form release agent prior to placing reinforcing. Wipe off excess with rags leaving the form surface thinly coated. Prevent over-spray on reinforcing steel which will destroy bond. Recoat forms after cleaning for re-use.

3.10 RE-USE OF FORMWORK

.1 Lumber, plywood and steel forms may be re-used after all nails have been withdrawn and surfaces to be in contact with concrete are thoroughly cleaned and repaired before re-use.

.2 Re-use of plywood for exposed work shall be limited to one re-use unless further re-use is approved by consultant.

.3 Board formwork shall not be re-used for exposed work.

3.11 REMOVAL OF FORMWORK

.1 Submit stripping and pouring schedule to the consultant for approval, prior to start of work.

.2 Notify the engineer of intention to strip forms in advance of removal. The minimum strength of concrete in place for safe removal of soffit forms for horizontal or inclined members shall be 70% of the specified 28-day strength, with the added provision that the stripped member shall be of sufficient strength to safely carry its own weight together with superimposed construction loads.

Cantilevers longer than 1.5 m shall obtain 100% of its specified strength before removal of soffit forms.

Strip wall forms before removal of shores beneath slabs or beams and strip slabs prior to removal of shores beneath beams.

.3 Unless otherwise shown on the drawings side forms for vertical members (walls) may be stripped as soon as the concrete is sufficiently strong to stand unsupported, but not before 3 days after concreting. If such forms are stripped before 7 days after concreting, the member faces must be treated with a curing agent that will seal the surface and prevent loss of moisture of the concrete surface. For supply and application of curing compound, see Section 03 30 00.

.4 Do not strip within half full bays of a construction joint until new concrete beyond the construction joint has reached 70% of its specified 28-day strength.

.5 Notwithstanding the above, under no circumstances shall forms and shoring be removed until concrete has gained sufficient strength to carry load and all possible construction and design loads liable to be imposed upon it. The Contractor shall be fully responsible for the safe removal of the forms and for any damage resulting from early removal of the forms including excessive deflections.

3.12 TOLERANCES, VERTICAL SURFACES

.1 All vertical lines shall be within 6 mm of plumb per storey, and not exceeding 12 mm of drift over the entire structure.

.2 Offset of continuous vertical elements in subsequent floors shall not exceed 6 mm.

.3 Relative position of vertical elements: Errors in any distance among the locations of faces of vertical elements shall not exceed 12 mm for adjacent members, 25 mm for any members.

3.13 TOLERANCES, HORIZONTAL SURFACES

.1 Variation of level (to be measured before removal of shores), shall not exceed:

1. 6 mm in 3 m.
2. 10 mm in 6 m.
3. 1:300 in one bay.

3.14 CONCRETE DIMENSIONS

.1 Errors in concrete dimensions for any member: shall not exceed 5 mm less or 10 mm more than true.

.2 The engineer shall have the right to have discrepancies corrected if they exceed the above limits.

END OF SECTION